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30 June 2014

DETAIL SPECIFICATION

FIXTURES, LIGHTING; AND ASSOCIATED PARTS; SHIPBOARD USE, GENERAL SPECIFICATION FOR

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This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers fluorescent, incandescent lighting and solid state lighting (SSL) (see 6.8.6) luminaires and associated parts for legacy replacement and new luminaire development fixtures (lights) and associated parts, used for detail and general illumination on naval ships and submarines.

1.2 Classification. Lighting fixtures are of the following types and classes, as specified (see 6.2).

1.2.1 Types.

a. Type I – Fluorescent.

b. Type II – Incandescent.

c. Type III – Solid state.

1.2.2 Classes.

a. Class 1 – Detail illumination.

b. Class 2 – General illumination.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-H28 - Screw-Thread Standards for Federal Services

COMMERCIAL ITEM DESCRIPTIONS

A-A-50552 - Fittings for Cable, Power, Electrical and Conduit, Metal, Flexible

A-A-59125 - Terminal Boards, Molded, Barrier Screw and Stud Types and Associated Accessories

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901 - Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for

MIL-DTL-917 - Electric Power Equipment, Basic Requirements for

MIL-DTL-5423 - Boots, Dust and Water Seal (for Toggle and Push-Button Switches, Circuit Breakers, and Rotary-Actuated Parts), General Specification for

MIL-DTL-15024 - Plates and Tags for Identification of Equipment, General Specification for

MIL-P-15024/5 - Plates, Identification

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MS16569	-	Switch, Toggle, Double Pole, Single Throw (for Use in Lighting Fixtures)
MS16656	-	Switch, Toggle, Single Pole, Single Throw
MIL-DTL-24191	-	Plastic Sheet, Cell or Continuous Cast, Acrylic, Shipboard Application (Illumination and Signal Lighting)
MIL-DTL-24640	-	Cables, Lightweight, Low Smoke, Electric, for Shipboard Use, General Specification for
MIL-DTL-24643	-	Cables, Electric, Low Smoke Halogen-Free, for Shipboard Use, General Specification for
MIL-DTL-24643/3	-	Cable, Electrical, -20 °C to +90 °C, 600 Volts, Types LSSHOF, LSDHOF, LSTHOF, and LSFHOF
MIL-DTL-24643/20	-	Cable, Electrical, -20 °C to +105 °C, 1000 Volts, Type LS7SGU
MIL-DTL-24643/26	-	Cable, Electrical, -20 °C to +150 °C, 600 Volts Types LSDPS, LSTPS, LSFPS, LS7PS, LSDPSN, LSTPSN, LSFPSN, and LS7PSN
MIL-PRF-24712	-	Coatings, Powder, Thermosetting
MIL-I-24768	-	Insulation, Plastics, Laminated, Thermosetting; General Specification for
MIL-I-24768/1	-	Insulation, Plastic, Laminated, Thermosetting, Glass-Cloth, Melamine-Resin (GME)

(See supplement 1 for list of specification sheets.)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-108	-	Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment
MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited)
MIL-STD-461	-	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-1399-300-1	-	Department of Defense Interface Standard Section 300, Part 1 Low Voltage Electric Power, Alternating Current
MIL-STD-1474	-	Noise Limits
MIL-STD-3009	-	Lighting, Aircraft, Night Vision Imaging System (NVIS) Compatible

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-217	-	Reliability Prediction of Electronic Equipment
MIL-HDBK-454	-	General Guidelines for Electronic Equipment

(Copies of these documents are available online at https://quicksearch.dla.mil.)

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2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NAS411-1 - Hazardous Materials Target List

(Copies of this document are available online at www.aia-aerospace.org.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASA S1.4 - Electroacoustics – Sound Level Meters – Part 1: Specifications

ANSI C78.180 - Specifications for Fluorescent Lamp Starters

ANSI C78.377 - Electric Lamps - Specifications for the Chromaticity of Solid-State Lighting Products

(Copies of these documents are available online at http://webstore.ansi.org/.)

AMERICAN WELDING SOCIETY (AWS)

WHB - Welding Handbook

(Copies of this document are available online at www.aws.org.)

ASME INTERNATIONAL

ASME B46.1 - Surface Texture (Surface Roughness, Waviness, and Lay)

ASME Y14.36 - Surface Texture Symbols

(Copies of these documents are available online at www.asme.org.)

ASTM INTERNATIONAL

ASTM A342/A342M - Standard Test Methods for Permeability of Weakly Magnetic Materials

ASTM D635 - Standard Test Method for Rate of Burning and/or Extent and Time of Burning of Plastics in a Horizontal Position

ASTM D2843 - Standard Test Method for Density of Smoke from the Burning or Decomposition of Plastics

ASTM D3955 - Standard Specification for Electrical Insulating Varnishes

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ASTM D4802	-	Standard Specification for Poly(Methyl Methacrylate) Acrylic Plastic Sheet
ASTM D5948	-	Standard Specification for Molding Compounds, Thermosetting
ASTM F1836M	-	Standard Specification for Stuffing Tubes, Nylon, and Packing Assemblies (Metric)

(Copies of these documents are available online at www.astm.org.)

ILLUMINATING ENGINEERING SOCIETY OF NORTH AMERICA (IES)

ANSI/IES LM-41	-	Approved Method: Photometric Testing of Indoor Fluorescent Luminaires
ANSI/IES LM-46	-	Approved Method: Photometric Testing of Indoor Luminaires Using High Intensity Discharge or Incandescent Filament Lamps
ANSI/IES LM-79	-	Approved Method: Optical and Electrical Measurements of Solid-State Lighting Products
ANSI/IES LM-80	-	Approved Method: Measuring Luminous Flux and Color Maintenance of LED Packages, Arrays, and Modules
ANSI/IES LM-84	-	Approved Method: Measuring Luminous Flux and Color Maintenance of LED Lamps, Light Engines, and Luminaires
ANSI/IES TM-21	-	Technical Memorandum: Projecting Long-Term Lumen, Photon, and Radiant Flux Maintenance of LED Light Sources
ANSI/IES TM-28	-	Technical Memorandum: Projecting Long-Term Luminous Flux Maintenance of LED Lamps and Luminaires

(Copies of these documents are available online at www.ies.org.)

SAE INTERNATIONAL

SAE AMS2175	-	Castings, Classification and Inspection of
SAE AMS-STD-595	-	Colors Used in Government Procurement
SAE ARP4392	-	Lighting, Aircraft Exterior, Night Vision Imaging System (NVIS) Compatible
SAE ARP5825	-	Design Requirements and Test Procedures for Dual Mode Exterior Lights

(Copies of these documents are available online at www.sae.org.)

UNDERWRITERS LABORATORIES INC. (UL)

UL 486A-486B	-	Wire Connectors
UL 486C	-	Splicing Wire Connectors
UL 496	-	Lampholders
UL 844	-	Standard for Luminaires for Use in Hazardous (Classified) Locations

(Copies of these documents are available online at www.comm-2000.com.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.3 Type III luminaires. Type III luminaires (see 6.8.5) may function as a replacement for fluorescent (type I) and incandescent (type II) lighting systems and shall meet the specifications identified in the applicable specification sheet. Type III luminaires are not limited to being replacements for type I and type II fixtures.

3.4 Materials.

3.4.1 General requirements. The materials used in the construction of the fixtures or parts shall be as specified herein and of a type, class, form, and grade that is readily available from normal sources of supply without the necessity for additional treatment or processing other than that which is normal to, or readily supplied by, the industry. Compliance of all materials with the minimum requirements and suitability of each material for its specific use and for the service intended shall be ascertained. Material requirements fall into the following general categories.

3.4.1.1 Specific materials. Where a specific material and material specification is specified herein or in the specification sheet, the material used shall conform to that specification. Materials purchased in accordance with the manufacturer's specification or specifications other than those indicated shall be considered acceptable only if materials and quality assurance procedures do, in fact, conform to the minimum requirements of the referenced specification.

3.4.1.2 Substitute materials. Material of equivalent grade may be substituted in lieu of the specified materials, provided that all other requirements of the individual specification are met and Technical Authority (see 6.8.9) approval or direction is obtained (see 6.7).

3.4.1.3 Material not specified. Where a specific material is not specified or the Technical Authority acceptance is not required, any material may be selected that will satisfactorily perform the intended function in the fixture, light, or part and will otherwise comply with specification requirements.

3.4.1.4 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4.2 Prohibited materials. Light fixtures shall not contain any chemicals categorized as "prohibited" in accordance with T9070-AL-DPC-020/077-2 for NAVSEA programs or NAS 411-1 for all other programs.

3.4.2.1 Prohibited materials in accordance with MIL-DTL-917. Materials prohibited by MIL-DTL-917 shall not be used. Flammable or explosive material, material which can produce toxic or suffocating fumes, or material which can produce a smoke density rating above 50 (see table I) when burned shall not be used unless specifically approved (see 6.7). Prohibited materials (toxic, flammable, fragile, radioactive, magnesium, mercury, carcinogens, and asbestos), when allowed for the specific use, shall conform to the requirement of MIL-DTL-917, except as specified herein.

3.4.2.2 Flammable plastic materials. As a guide, plastic materials, when not specified herein or in the applicable specification sheet, shall be selected as specified in table I.

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TABLE I. Flammability characteristics of plastics.

Characteristic	Specimen (Inches)	Applicable Test Method	Limit
Flammability rate	5L × ½W × ¼T	ASTM D635	See the material resistance burning requirements in MIL-DTL-917.
Smoke density rating	1L × 1W × ¼T	ASTM D2843	50 maximum

3.4.3 Fungus-inert materials. Fungus-inert materials shall meet rating 0 or 1 of MIL-STD-810, method 508, when tested for a period of 28 days in accordance with MIL-STD-810, method 508. See guideline 4 of MIL-HDBK-454 for additional information.

3.4.4 Arc-resistant materials. Arc-resistant materials shall be in accordance with MIL-DTL-917. See guideline 26 of MIL-HDBK-454 for additional information.

3.4.5 Metals. Metals shall be selected or processed and applied in a manner that provides corrosion resistance. Metals that are not inherently corrosion-resistant (see 3.4.5.2) shall be processed (treated, plated, or painted) to provide corrosion resistance.

3.4.5.1 Selection of metals in direct contact. In order to minimize corrosion attack due to electrolytic action between dissimilar metals in contact with each other, metal-to-metal contacts shall be limited to those metals which, when coupled, are in accordance with MIL-DTL-917.

3.4.5.2 Corrosion-resistant. For a list of commonly used corrosion-resistant metals, see MIL-DTL-917.

3.4.5.3 Aluminum. Aluminum alloy shall be used insofar as practicable. Aluminum alloys, except castings, shall conform to ASTM standards. Aluminum alloy casting shall be the manufacturer's choice, except sand castings and permanent mold castings shall be in accordance with class 2B of SAE AMS2175. Grade shall be the manufacturer's choice.

3.4.5.4 Nonferrous material. Nonferrous materials, except aluminum, shall conform to commercial standards.

3.4.6 Plastics. Plastic materials shall be in accordance with the requirements of MIL-DTL-917, 3.4.2.2 of this specification, and as specified herein.

3.4.6.1 Uniformity. Plastic materials shall be free of defects such as blemishes, embedded particles, bubbles, scratches, and striations.

3.4.6.2 Mechanical parts. Whenever plastic materials are considered for mechanical applications as replacements or substitutes for metal parts, they shall meet the flammability characteristics specified in 3.4.2.2.

3.4.6.3 Thermoplastic materials. Plastics that melt or soften under test conditions imposed by the applicable specification sheet shall not be used.

3.4.6.4 Electrical insulating parts. Plastics used for electrical insulating parts shall be of the type specified herein.

3.4.6.4.1 Plastics, laminated. Laminated plastic material shall be a glass cloth bonded with melamine resin in accordance with MIL-I-24768 and MIL-I-24768/1. No other laminated plastic material shall be used for electrical insulation. All cut surfaces of laminated plastic shall be given two coats of varnish in accordance with ASTM D3955 to prevent absorption of moisture or a conformal coating shall be used.

3.4.6.4.2 Thermosetting, molded. Thermosetting, molded insulated parts shall conform to type MAI-60 or type MMI-30 of ASTM D5948.

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3.4.7 Springs. Material for springs, retaining clips, and retaining rings shall be as specified in the applicable specification sheet. See the springs guideline in MIL-HDBK-454 for additional information in regard to common design and material decisions.

3.4.8 Magnetic permeability. The magnetic permeability of all surfaces of completed fixtures shall be 2.0 or less after fabrication when tested as specified in 4.8.18. Unless otherwise specified (see 3.1 and 6.2), the magnetic permeability requirements shall not apply to shock mounts and fastening devices, such as rivets and screws.

3.5 Parts and features, mechanical.

3.5.1 Hardware. Unless otherwise specified in the applicable specification sheet, hardware (bolts, nuts, screws, washers, and miscellaneous hardware) shall be of a good commercial grade material compatible with that of the basic fixture. Steel (corrosion-resistant steel excepted) hardware shall be zinc plated.

3.5.2 Dimensions and tolerances. Unless otherwise specified herein, the dimensions shown on the figures in this specification are in inches. For associated specification sheets, dimensions shall be as shown in the applicable specification sheet. Unless otherwise specified in the applicable specification sheet, the following tolerances shall apply:

a. Fractional dimensions: ±1/64 inch

b. Decimal dimensions: ±0.005 inch

c. Angular dimensions: ±0 degree, 15 minutes

Unless otherwise specified in the applicable specification sheet, a tolerance of ±1/32 inch is acceptable on fractional dimensions that are controlled by welding, brazing, and shock mounts. This wider tolerance shall not interfere with the interchangeability of assemblies or parts.

3.5.3 Assembly fabrication. Assembled fixtures shall have no loose parts, subassemblies, or baffles that will rattle or detract from their quiet operation during vibration tests, or from transmitted vibration under operating conditions of the ship. Auxiliary dampening materials shall not be used for cushioning parts or wiring to achieve acceptable quiet operation. Rivets, pads, brackets, assemblies, and subassemblies shall be securely attached to prevent rotation, rattling, turning, or bending, unless the rotation or bending is part of the design. No loosening of these items shall be evident at the conclusion of the shock and vibration tests specified in 4.8.7 and 4.8.8.

3.5.4 Drilling, countersinking, and tapping. Drilling, countersinking, and tapping shall be done before plating or finish is applied. Tapped holes that are used for normal replacement of parts shall be countersunk.

3.5.5 Sharp edges. Sharp edges and corners that are accessible shall be given a slight radius.

3.5.6 Castings and molded parts. Castings shall be free from cold shuts, blow holes, or any imperfections that may affect strength. Surfaces of castings shall have burrs and fins removed. Flash shall be removed from molded parts.

3.5.7 Surface textures. Surface textures specified in the applicable specification sheet shall be in accordance with ASME B46.1. Surface texture symbols shall be in accordance with ASME Y14.36.

3.5.8 Threaded parts and devices.

3.5.8.1 Threads. Threads for threaded fastening devices shall be in accordance with FED-STD-H28. The threads shall be right-hand, coarse-thread series, unified thread form, class 2A or 2B or American National thread form, class 2. Other thread series and classes, such as fine thread, may be used where it is necessary to ensure functional operation of the equipment. Threads shall be checked during production run with “GO” and “NO-GO” gauges to ensure conformance to FED-STD-H28.

3.5.8.2 Engagement of threaded parts. Threaded parts shall engage by at least four full threads in soft metals like aluminum and aluminum alloys. A minimum of three full threads shall be used in harder materials such as brass or steel. When a screw mates with a plastic part, a threaded metal insert shall be molded therein (pressed-in inserts are not acceptable), except as specified in 3.5.8.4.

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3.5.8.3 Thread cutting screws. Thread cutting screws (sheet metal screws and self-tapping screws that remove material while cutting the thread) shall not be used. Thread forming screws (self-tapping screws that compress the parent material around the engagement hole while forming the thread) are acceptable.

3.5.8.4 Threads in plastics. Threads shall not be employed in plastic parts without the use of threaded metallic inserts, except in instances where the use of the inserts would adversely affect the electrical or mechanical characteristics of the part. Inserts shall be secured by knurling or other method that will prevent their movement in the plastic material.

3.5.8.5 Threads in aluminum. In general, threads in aluminum alloys shall not be used. Wherever practical, through bolt holes shall be used. Inserts in aluminum shall be provided only where screws or bolts will be removed for routine maintenance of the equipment or where maximum stress in the screw or bolt is required for alignment of a vital part. Inserts in aluminum shall not be provided for securing identification or information plates, terminal boards, or other items that are removed only when equipment is modified. Inserts shall be of austenitic, corrosion-resistant steel.

3.5.8.6 Anti-seizure coating. Aluminum and steel fastening parts (screws and bolts) in contact with threaded aluminum shall be coated with an anti-seize compound.

3.5.8.7 Threads locking. Unless otherwise specified herein, nuts, bolts, studs, and screws shall be secured by a locking device. Satisfactory locking devices are split ring-type lock washers, nut and lock nut, self-locking nuts, and castellated nuts with cotter pin or safety wiring. Internal tooth-type lock washers shall not be used. External tooth-type lock washers may be used where the weight of the part does not exceed 6 ounces per screw. Nuts or screws with captive lock washers may be used, provided they are replaceable with common-type screws and lock washers. Oval head screws, when used with cup-type washers, and pan head screws on identification plates do not require locking devices. Thread forming screws do not require locking devices. Screws with an O-ring seal do not require locking devices. When a bonding agent is used to permanently secure two or more threaded fastening devices, an oversized thread may be used to provide a stronger bond. The oversized threads shall be checked during the production run with “GO” and “NO-GO” gauges to ensure conformance to the original established dimensional limits.

3.5.9 Seals and tamper-resistant screws for explosionproof and combined explosionproof-watertight portable fixtures. Lead seals or tamper-resistant screws shall be used for explosionproof and combined explosionproof-watertight portable fixtures. Seals or screws that require the use of special tools shall be approved from the Technical Authority. See 3.10.1, 6.8.7, and MIL-DTL-917 for requirements for special tools.

3.5.10 Gaskets. Unless otherwise specified in the applicable specification sheet, gaskets shall be as specified in 3.5.10.1 through 3.5.10.5.

3.5.10.1 Window gaskets for type I, class 2 fixtures and type III luminaires using legacy fluorescent windows. Window gaskets for type I, class 2 fixtures and type III luminaires utilizing legacy type I, class 2 windows shall be neoprene tubing (45-55 durometer, type A hardness), 0.295-inch maximum and 0.245-inch minimum outside diameter (od), and 0.062-inch inside diameter (id). No petroleum or silicon-based lubricants shall be used when extruding gaskets. Gaskets shall be securely cemented at the scarf joint and shall completely fill the groove in the luminaire without stretching or bunching. Gaskets shall be retained in their grooves without resorting to cement. Retainer may be continuous (see 3.5.11.7).

3.5.10.2 O-ring gaskets. Unless otherwise specified in the applicable specification sheet, O-ring gaskets shall be commercial neoprene (65-75 durometer, type A hardness). No petroleum- or silicon-based lubricants shall be used when molding gaskets. Sizes of O-ring gaskets for type I legacy window assemblies shall be as shown on figures 1, 2, and 4.

3.5.10.3 U-shaped gaskets. U-shaped gaskets for parabolic aluminized reflector (PAR) lamps (see 6.8.3), roundels, and lenses shall be commercial neoprene rubber (45-55 durometer, type A hardness) of a continuous band and of a circumference less than the diameter of the PAR lamp, roundel, or lens. No petroleum- or silicon-based lubricants shall be used when molding gaskets. The gaskets shall be so arranged that when stretched over the PAR lamp, roundel, or lens, the material will form over the edge and the faces of the PAR lamp, roundel, or lens to form a U-shaped gasket.

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3.5.10.4 Gasket coating. O-rings and tubular gaskets shall be coated with a glycerin lubricant before final assembly in the fixtures.

3.5.10.5 Window gaskets for type III luminaires. Window gaskets for type III luminaires shall be to the manufacturer's design when not utilizing legacy window components. No petroleum- or silicon-based lubricants shall be used when extruding gaskets. Gaskets shall be securely cemented at the scarf joint and shall completely fill the groove in the luminaire without stretching or bunching. Gaskets shall be retained in their grooves without resorting to cement. Retainer may be continuous (see 3.5.11.7).

3.5.11 Enclosures. Enclosures shall be as specified in 3.5.11.1 through 3.5.11.7, and as specified in the applicable specification sheet. There shall be no evidence of failure when fixtures and parts are tested as specified in 4.8.14 to determine effectiveness of the enclosures.

3.5.11.1 Enclosures, type I, class 1. Enclosures for type 1, class 1 fixtures shall be as specified in the applicable specification sheet. Unless otherwise specified in the applicable specification sheet, thickness of aluminum, where used, shall be not less than 0.090 inch. Grade of materials where not specified shall be the manufacturer's choice.

3.5.11.2 Enclosures, type I, class 2. Enclosures for type 1, class 2 fixtures shall be as specified in the applicable specification sheet. The enclosures shall be made of aluminum. Aluminum or zinc diecast ends may be used in assembly with sheet aluminum midsection. Die cast ends shall not be in direct contact with the shock mounts. Grade of aluminum used shall be the manufacturer's choice. Unless otherwise specified in the applicable specification sheet, thickness of aluminum shall be not less than 0.090 inch. For personnel safety while cleaning or relamping fixture, a small compartment with removable cover shall be provided at each end within the enclosure for the purpose of containing the necessary parts and wiring connections. Areas in way of shockmount bolts shall be indented to receive gaskets and shock mounts as shown on figure 6. The enclosure shall be sufficiently rigid to ensure effective alignment and closure between enclosure and lens upon completion of shock tests conducted on assembled fixtures. Cable entrance hole shall be of a size and shape to suit the style of cable entrance fitting. Protective cover shall be provided as specified on figure 19.

3.5.11.3 Enclosures, type II, classes 1 and 2. Enclosures for type II, classes 1 and 2 fixtures and lights shall be as specified in the applicable specification sheet.

3.5.11.4 Enclosures, type III. Enclosures for type III, classes 1 and 2 luminaires, when designed as a replacement for legacy luminaires, shall mount up in the same locations using the same hardware as the legacy fixtures and shall not exceed the dimensional envelope of the legacy fixtures. Grade of materials where not specified shall be the manufacturer's choice.

3.5.11.4.1 Additional class 2 requirements. The enclosure shall be sufficiently rigid to ensure effective alignment and closure between enclosure and lens upon completion of shock tests conducted on assembled fixtures. Cable entrance hole shall be of a size and shape to suit the style of cable entrance fitting.

3.5.11.5 Parts. Enclosures shall incorporate all parts specified in the applicable specification sheet. Parts shall be firmly attached to the enclosure.

3.5.11.6 Drainage. When specified in the applicable specification sheet, enclosures shall provide for drainage of water vapor condensation through drain holes or other means located so as to provide for maximum drainage when fixtures and lights are mounted in their normal operating position.

3.5.11.7 Gasket groove. Gasket groove for watertight type I, class 2 fixtures and type III luminaires utilizing legacy type I, class 2 windows shall be as shown on figure 7.

3.5.12 Shock. Fixtures and parts (as specified in 3.5.12.1 through 3.5.12.3) shall withstand, without damage or loosening of parts, the shock test specified in 4.8.7.

3.5.12.1 Types I, II, and III, class 1 fixtures. Unless otherwise specified in the applicable specification sheet, types I, II, and III, class 1 fixtures shall withstand the high-impact (H.I.) shock for grade B, type A equipment in accordance with MIL-DTL-901.

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3.5.12.2 Types I, II, and III, class 2 fixtures. Unless otherwise specified in the applicable specification sheet, types I, II, and III, class 2 fixtures shall withstand the H.I. shock for grade A, type A equipment in accordance with MIL-DTL-901. Type I lamps shall show no permanent loss of illumination during shock test. Failure of the type II lamp filament does not constitute failure of the fixture tested. Type III luminaires shall show no permanent loss of illumination during test.

3.5.12.3 Parts. When specified in the applicable specification sheet, parts shall withstand the shock test specified for the basic fixture in which they are used (see 3.5.12.1 and 3.5.12.2).

3.5.13 Vibration. When specified in the applicable specification sheet, and for all type III luminaires, fixtures and parts shall withstand, without damage or loosening of parts, the type I vibration test in accordance with MIL-STD-167-1, as specified in 4.8.8. Use of locking devices is satisfactory to prevent loosening of parts. Type I lamps shall show no permanent loss of illumination during the vibration test. Failure of type II lamp filament does not constitute failure of fixture tested. Type III luminaires shall show no permanent loss of illumination during test.

3.5.14 Noise. When specified in the applicable specification sheet, and for all type III luminaires, fixtures, lights, or parts shall not exceed the sound pressure level of 28 decibels (dB) referenced to 0.0002 microbar or the sound pressure level specified in the applicable specification sheet for airborne noise, only when tested as specified in 4.8.9.

3.5.15 Salt fog. Fixtures and parts made of steel or aluminum for weather deck, well deck, or hangar bay installation shall withstand the salt fog test specified in 4.8.10 without excessive corrosion (see 6.8.2).

3.5.16 Shock mounts. Shock mounts shall be furnished as specified in the applicable specification sheet. Thickness of material shall be the manufacturer's choice if failure occurs under the shock and vibration tests. Type III luminaires intended as direct replacements for type I and type II luminaires shall not be required to utilize shock mounts if testing can be passed without them.

3.5.17 Ambient temperature. The fixtures, lights, and parts shall operate in the temperature specified in the applicable specification sheet when tested as specified in 4.8.19. Unless otherwise specified (see 6.2), all type III luminaires shall be designed to operate within an ambient temperature range of -13 to 140 °F (-25 to 60 °C).

3.6 Parts and features – electrical.

3.6.1 Safety. Fixtures and parts shall be constructed in a way that will ensure safety to operating and maintenance personnel. When properly installed and the enclosure is grounded, there shall be no accessible way for operating personnel to receive an electric shock, even though an internal fault may exist between two circuits, between any circuit and a structural member, or between any circuit and ground. The design shall hold to a practical minimum the possibility of maintenance personnel being exposed to electric shock while servicing, adjusting, or checking out the fixtures or parts.

3.6.2 Lamps in fixtures. Unless otherwise specified (see 6.2), lamps for type II fixtures shall not be furnished. Lamps for type I fixtures shall be as specified in the applicable specification sheet and shall be furnished assembled in the fixture. Type III fixtures shall be furnished fully assembled.

3.6.3 Lamplocks for type I fixtures. Lamplocks for type I fixtures shall be as specified in the applicable specification sheet and shall be furnished assembled in the fixture. Lamplocks shall provide for maximum protection against lamp loosening or rotation under the shock and vibration tests.

3.6.4 Lampholders. Lampholders shall be in accordance with the naval-use lampholders section of UL 496 and as specified herein. The type shall be as specified in the applicable specification sheet.

3.6.4.1 Lampholder spacing for type I fixtures and lights. Unless otherwise specified in the applicable specification sheet, spacing between faces of lampholder supports at the point of lampholder attachment for type I fixtures and lights shall be as shown on figure 21.

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3.6.4.2 Insulating spacer for type II fixtures and lights. An insulating spacer shall be installed between the lampholder metal mounting base and the lampholder in the case where the lampholder is provided with screws or other devices for electrical connections that are exposed to the mounting base. Material shall be as specified in 3.4.6.4.

3.6.4.3 Counterbored holes. Counterbored holes in the bottom of lampholders or insulated bases shall not be filled with sealing wax or rosin after screws are in place.

3.6.5 Starter sockets for type I fixtures and lights. Starter sockets for type I fixtures and lights shall be in accordance with the naval-use lampholders section of UL 496. The type shall be as specified in the applicable specification sheet.

3.6.6 Starters for type I fixtures and lights. When specified in the applicable specification sheet, starters for type I fixtures and lights shall be as specified in table II. Unless otherwise specified (see 6.2), the type to be furnished shall be as specified in the applicable specification sheet and shall be furnished assembled in the fixture or light. The interior of starters shall be filled with wax in cases where used in fluorescent lights having explosionproof requirements. Starters shall be furnished with a ceramic condenser to reduce starting electromagnetic noise and with a paper sleeve inside the starter can to reduce noise from vibration of interior parts against can. All starters shall have a metallic enclosure.

TABLE II. Starters.

Voltage	Description	Fluorescent Lamp	
		6–8 Watts	15–20 Watts
Alternating current	Item name	Simple glow switch, without lockout	Simple glow switch, without lockout
	Specification	ANSI C78.180	ANSI C78.180
	Commercial designation	FS-5	FS-2
	National Stock Number	6250-00-299-5962	6250-00-299-2884

3.6.7 Starter washers for type I fixtures and lights. Starter washers as specified on figure 5 shall be provided for type I, classes 1 and 2 fixtures and lights that are furnished with starters (see 3.6.6). The washer shall be slipped over contacts of the starter. The washer and starter shall then be inserted into the starter socket and rotated to lock in the normal manner to provide a snug, rattle-free fit.

3.6.8 Ballasts for type I fixtures and lights. Ballasts for type I fixtures and lights shall be as specified in the applicable specification sheet and herein.

3.6.8.1 Mounting. Ballasts shall be mounted so that they will not become loose under the shock and vibration tests. Ballast mounting shall make good contact with the fixture housing so as to aid conduction of normal operating heat away from the ballast to reduce noise. Ballasts shall be secured only by bolts (machine screws) and shall be capable of ready removal for replacement.

3.6.9 Terminal boards. Unless otherwise specified in the applicable specification sheet, terminal boards shall be in accordance with A-A-59125. The type shall be as specified in the applicable specification sheet.

3.6.9.1 Mounting. Terminal boards for receiving ship's cable shall be secured to a fixed portion of the enclosure, not to movable assemblies that may require flexing of the ship's wiring during normal operation, maintenance, or servicing. Terminal boards shall be mounted so that they will not be broken or stressed by distortion of the enclosure. Terminal boards shall be secured only by bolts (machine screws) and shall be capable of ready removal and replacement.

3.6.10 Switches. When specified (see 6.2), switches shall be furnished as specified in the applicable specification sheet and as specified herein.

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3.6.10.1 Toggle switches. Double-pole, single-throw switches shall conform to MS16569. Single-pole, single-throw switches shall conform to MS16656.

3.6.10.2 Push-button switches. Push-button switches shall be as specified in the applicable specification sheet.

3.6.10.3 Boots. Rubber boots in accordance with MIL-DTL-5423 shall be used to achieve the degree of enclosure effectiveness specified in the applicable specification sheet. Unless otherwise specified in the applicable specification sheet, type and style of boot shall be the manufacturer's choice.

3.6.11 Wiring. Necessary internal wiring shall be furnished and installed. Connections at screw terminals shall be made with pressure grip (solderless) connectors, in accordance with UL 486A-486B. Splice connections shall be kept to a minimum and shall be made with UL-approved insulated connectors, either crimp- or screw-type. Wires parallel to lamps in type I fixtures shall be run in a conduit or shall be confined between reflector and housing. Except for terminal board connections, there shall be no exposed uninsulated electrical terminal joints, junctions, wiring, or wiring accessories. Polyvinyl chloride insulation shall not be used. Wiring for type I fixtures shall be as specified in the applicable specification sheet and figures 22, 23, and 24.

3.6.11.1 Electrical connections for type III luminaires. Unless otherwise specified (see 6.2), connections between the solid-state device and driver circuits shall be the quick-disconnect plug-in type with built-in locking mechanism. Incoming power circuit conductors and internal wiring conductors shall terminate on terminal boards or into non-permanent, reusable wire connectors that allow for the physical removal of each conductor from the connector without cutting the conductor. Terminal boards, lugs, and circuit boards shall comply with the requirements of MIL-DTL-917. Non-permanent wire connectors shall meet the requirements of UL 486A-486B or UL-486C. Wire nuts and closed-end compression connectors are not permitted within the luminaire enclosure.

3.6.11.2 Hook-up wires. Wires shall be of stranded, soft annealed copper with a cross section large enough to provide ample and safe current carrying capacity and mechanical strength.

3.6.11.3 Harnessing. Wiring shall be neatly formed into groups which shall be laced, tied, taped, or clamped in a manner that provides support and prevents chafing of the wire insulation. Requirements for harnessing listed in MIL-DTL-917 should be used as guidance.

3.6.11.4 Insulation protection. Where wires are run through holes in metal partitions, these holes shall be furnished with grommets for the mechanical protection of insulation that otherwise could be subject to abrasion. Care shall be exercised in the running of wires to ensure that they are not carried over or bent around any sharp corner or edge.

3.6.12 Cable and cable entrances. Cable shall be as specified in the applicable specification sheet. Unless otherwise specified in the applicable specification sheet, cables for types I, II, and III, classes 1 and 2 fixtures, shall be 7-conductor and low smoke in accordance with MIL-DTL-24640 or MIL-DTL-24643. Cable length shall be measured from the cable entrance of fixture, light, or part to the end of the cable. A cable length tolerance of ±3 inches shall apply. The following stuffing tubes or box connectors shall be used as cable entrance:

a. Unless otherwise specified in the applicable specification sheet, stuffing tubes, stuffing tube assemblies, and stuffing tube parts shall be in accordance with ASTM F1836M. Holes for stuffing tubes shall be smooth and true without any distortion of the box. Packing assembly and packing nut in accordance with ASTM F1836M shall be used when cable entrances are integral with the housing.

b. Box connectors shall be type I in accordance with A-A-50552. Grade and style shall be as specified in the applicable specification sheet. Box connectors shall not be used in watertight fixtures or parts.

3.6.13 Ground potential and grounding.

3.6.13.1 Electrical circuits. Fixtures and parts shall operate satisfactorily on an ungrounded ship's power system. The design shall be such that it does not impose a ground upon the electrical power system from which it is energized.

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3.6.13.2 Exposed metal (or other conductive) parts. Construction of fixtures and parts shall be such that all exposed parts of metal or other electrically conductive material are at ground (ship's hull) potential at all times. Exposed metal portions of electrical parts (switches and lampholders) or other parts located near electrical circuits (including parts inside enclosures where access is required for operation or adjustment) shall be in intimate physical contact with the frame of the equipment or electrically connected to the frame if these parts could touch the electrical circuits as a result of deformation, wear, or insulation failure.

3.6.13.3 Ground connection. The supply cable to fixtures shall be provided with a grounding conductor (green or red) which terminates within the fixture. The ground conductor connection point shall be securely attached and electrically connected to the metallic or conducting structure most likely to contact the body. This shall be accomplished in such a manner that all exposed metallic or conducting parts make electrical contact with the grounding conductor when tested as specified in 4.8.11.

3.6.13.4 Continuity of grounding. Fixtures and parts provided with shock mounts or removable mounting plates shall be not greater than 0.10 ohm impedance between protruding tip of shock mount or mounting plate and any remote internal metallic structure of housing assembly when tested as specified in 4.8.12. Shock mounts and mounting plates shall not be painted.

3.6.13.5 Portable fixtures. The leakage current (see 6.8.4) of the portable fixtures shall be not greater than 5 milliamperes. Leakage current shall be measured as specified in 4.8.13.

3.6.14 Electromagnetic interference (types I and III only). Unless otherwise specified (see 6.2 and 6.6), electromagnetic interference requirements for types I and III fixtures shall conform to the requirements of MIL-STD-461 when tested as specified in 4.8.16.

3.6.15 Dielectric withstanding voltage. There shall be no evidence of breakdown, arcing, corona (audible or visible), or punctured insulation when the interior of the fixture or part is subjected to the test specified in 4.8.2.

3.6.16 Insulation resistance. The insulation resistance between all current-carrying and noncurrent-carrying parts of fixtures or parts shall be not less than 1 megohm when tested as specified in 4.8.3.

3.6.17 Fail-safe circuit design for type III luminaires. Light emitting diode (LED) circuits utilizing more than one LED shall be designed to ensure multiple open circuit, or failed LEDs shall not diminish electrical energy to other LEDs. This may be accomplished using bypass or alternative path circuits for series LEDs, or by adjustment of the electrical energy supplied to remaining LEDs wired in parallel to the failed LED (see 4.8.22).

3.6.18 Power interface requirements. Unless otherwise specified (see 6.2), all fixtures energized from alternating current sources shall be designed to operate on type I power as described in MIL-STD-1399-300-1 (see 4.8.21).

3.6.18.1 Input current waveform for types I and III fixtures. Unless otherwise specified (see 6.2),the MIL-STD-1399-300-1 limit for equipment less than 1 kilovolt ampere is not applicable to input current waveform testing, unless the fixture has less than 6 watts of input power.

3.6.19 Class 2 emergency light fixtures. Class 2 emergency light fixtures shall utilize a battery to provide useful light upon the loss of normal power. The battery backup system shall include charging circuitry, an input power sensing circuit, low battery voltage cutoff circuitry to maximize battery life, and devices for protection against high voltage spikes. Upon loss of input power, the system shall switch to battery power within 0.5 second. When input power is restored, the light shall return to its normal operating parameters and begin recharging the battery without any action from operators or personnel.

3.6.19.1 Class 2 emergency light fixture output life. Emergency light fixtures shall provide usable light for a minimum of 5 hours on a fully charged battery. Lumen output levels shall be as specified in 3.7.1.1.

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3.6.19.2 Class 2 emergency light fixture battery. The battery or battery pack (containing the battery and all necessary circuitry for charging and input power sensing) for an emergency light fixture shall be field replaceable. Connections between the battery or battery pack and the other circuitry of the light fixture shall be made with quick-disconnect connectors or other non-permanent splice connectors meeting the requirements of UL 486C. Unless otherwise specified in the applicable specification sheet, battery chemistry shall be approved separately by the Technical Authority.

3.6.19.3 Class 2 emergency light fixture charging system. The charging system for the emergency battery shall take a maximum of 20 hours to fully charge the battery and shall draw no more than 4 watts of power during any point of the recharge cycle. Battery charging performance shall be tested as specified in 4.8.23.

3.7 Light output and associated hardware.

3.7.1 Candlepower distribution for fixtures. Candlepower distribution curves shall be symmetrical and the candlepower value shall be not less than 90 percent or more than 120 percent of the values in the curves shown in the applicable specification sheet. For angles above 80 degrees that have a candlepower of 50 or less, the +20/-10 percent rule does not apply. Tests shall be conducted as specified in 4.8.6.

3.7.1.1 Lumen output for emergency class 2 lighting. Light fixtures designed to run on an internal battery backup after loss of power shall provide a minimum of 240 lumens initially and no less than 140 lumens after 5 hours of battery runtime. Tests shall be conducted as specified in 4.8.6.8.

3.7.2 Footcandle distribution for fixtures. Footcandle distribution curves shall be symmetrical and the footcandle value shall be not less than 90 percent or more than 120 percent of the values in the curves as shown in the applicable specification sheet. Tests shall be conducted as specified in 4.8.6.

3.7.3 Color (chromaticity) and luminous transmittance for parts. The chromaticity in CIE coordinates and luminous transmittance for parts such as windows, globes, roundels, lenses, and filters shall be as specified in the applicable specification sheet and as specified herein. Color and luminous transmittance of plasticware parts shall be in accordance with MIL-DTL-24191. Color and luminous transmittance requirements shall be met prior to application of any diffusing process. If stippling is used for diffusing, the stippled surface shall be ground down and polished to a smooth surface or a test specimen shall be molded to a smooth surface of a nominal thickness as shown in the specification sheet for the part to be tested.

3.7.3.1 Red transparent. Red transparent color shall fall within the area bounded by the spectrum locus and $y = 0.335$, $x = 0.985-y$, as shown on figure 8. Red glassware shall have a light transmission value that follows the curves shown on figure 9 when specimens are tested as specified in 4.8.6.

3.7.3.2 Amber (yellow) transparent. Yellow transparent color shall fall within the areas bounded by the spectrum locus and

$y = 0.382$ $x = 0.454 + 0.240y$

$y = 0.790 - 0.667x$

as shown on figure 8. Luminous transmittance, using CIE illuminant A, shall be not less than 50 percent when specimens are tested as specified in 4.8.6.

3.7.3.3 Green transparent. Green transparent color shall fall within the area bounded by the spectrum locus and

$y = 0.390 - 0.171x$ $x = 0.360 - 0.080y$

$x = 0.650y - 0.030$

as shown on figure 8. Luminous transmittance, using CIE illuminant A, shall be not less than 20 percent when specimens are tested as specified in 4.8.6.

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3.7.3.4 Blue transparent. Blue transparent color shall fall within the area bounded by the spectrum locus and

y = 0.230 x = 0.133 + 0.600y

x = 0.400 - y

as shown on figure 8. Luminous transmittance, using CIE illuminant A, shall be not less than 2 percent when specimens are tested as specified in 4.8.6.

3.7.3.5 White translucent. White translucent, when tested as specified in 4.8.6, shall have luminous transmittance as specified in the applicable specification sheet. Transmitted light shall be distributed uniformly through the plastic.

3.7.3.6 Red translucent. Red translucent color shall fall within the area bounded by the spectrum locus and y = 0.335, x = 0.985-y, as shown on figure 8. Red material shall have a light transmission value that follows the curve shown on figure 9 when tested as specified in 4.8.6.

3.7.3.7 Clear (colorless) transparent. The light transmitted by clear transparent ware shall not be noticeably different in chromaticity from that of the illuminant. Luminous transmittance shall be not less than the value specified in the applicable specification sheet. Clear transparent ware shall transmit all light between wavelength 420 nanometers (nm) (violet) and 680 nm (red). Selective absorption at any wavelength shall not exceed 5 percent of the entire energy at that wavelength of the light source. Total absorption shall not exceed 18 percent.

3.7.4 Chromaticity for type III luminaires. Chromaticity is defined by (x) (y) coordinates on the CIE 1931 international chromaticity diagram. Areas within the curve for specified colors of light other than white are defined in 3.7.4.1 through 3.7.4.5, using four reference points and the spectrum locus to create a boundary (see figure 8 and 4.8.6.5).

3.7.4.1 White light. White light is defined as any light that has measurable (x) (y) coordinates in the CIE 1931 color space that falls anywhere within the total region identified in ANSI C78.377.

3.7.4.1.1 Correlated color temperature (CCT). The CCT and distance from the Planckian locus (D_{UV}) for white light shall fall within the tolerances defined by the Flexible CCT formula of ANSI C78.377. Unless otherwise specified (see 6.2), the target color temperature is 4,100K.

3.7.4.1.2 Color rendition. Color rendition for type III fixtures shall be determined using IES TM-30 as a guide. The Fidelity Index (R_f) shall be greater than 80. The Relative Gamut Index (R_g) shall be between 97 and 110 (see 4.8.6.7).

3.7.4.2 Red light. Chromaticity shall be within the area bounded by the spectrum locus and the following:

x = 0.6482 x = 0.7255

y = 0.3350 y = 0.2595

x = 0.6658 x = 0.7347

y = 0.3350 y = 0.2655

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3.7.4.3 Amber (yellow) light. Chromaticity shall be within the area bounded by the spectrum locus and the following:

x = 0.5510	x = 0.6171
y = 0.4143	y = 0.3820
x = 0.5548	x = 0.6080
y = 0.4544	y = 0.3820

3.7.4.4 Blue light. Chromaticity shall be within the area bounded by the spectrum locus and the following:

x = 0.0593	x = 0.2369
y = 0.2300	y = 0.1631
x = 0.1701	x = 0.1483
y = 0.2300	y = 0.0256

3.7.4.5 Night vision device (NVD)-friendly cyan light. NVD-friendly cyan chromaticity shall be within the area bounded by the spectrum locus and the following:

x = 0.0687	x = 0.2030
y = 0.2000	y = 0.3560
x = 0.1623	x = 0.0280
y = 0.2000	y = 0.3850

3.7.4.5.1 Night vision radiant intensity (NRI). The NRI values defined in SAE ARP4392 are equal to the unscaled Night Vision Radiance (NR) multiplied by the square of the distance from the light source to the measurement plaque. The NRI_B value of the luminaire shall be less than or equal to 5.0 E 3 Watts/steradian at full output. NRI_B is NRI based on the response of class B Night Vision Imaging System as defined in MIL-STD-3009.

3.7.5 Brightness for type III luminaires. Brightness (luminance intensity) shall not exceed the following limits. The brightness requirement is applicable to only type III, class 2 luminaires (see 4.8.6.4).

3.7.5.1 Brightness at nadir. Brightness shall not exceed 15,000 candela per square meter for the entire illuminated area for the vertical angle of 0 (see 4.8.6.4).

3.7.5.2 Brightness at angle. Brightness shall not exceed 8,000 candela per square meter for the entire illuminated area for the vertical angles of 45 degrees and above (see 4.8.6.4).

3.7.5.3 Local brightness. The ratio of the maximum luminance (luminance of the brightest 1-inch diameter) to the average luminaire luminance determined in 3.7.5.2 shall not exceed 5:1 at angles of 45, 55, 65, 75, and 85 degrees from nadir for crosswise and lengthwise viewing (see 4.8.6.4).

3.7.6 Windows and window assemblies. Windows and window assemblies shall be as specified in the applicable specification sheet and herein.

3.7.6.1 Windows and window assemblies for type I fixtures. Finish shall be as specified in 3.8.4.1. Annealing shall be as specified in 3.8.5.3. Typical styles of window assemblies are shown on figure 10.

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3.7.6.1.1 Material for windows. Material for windows shall be cast, or ASTM D4802 melt process category M-1, acrylic plastic sheet in accordance with MIL-DTL-24191 of a type and grade as specified in the applicable specification sheet. The requirements of MIL-DTL-24191 shall apply to the flat material prior to forming or fabrication. Windows after forming or fabrication shall be as specified in the applicable specification sheet. Injection molded, or extruded material, is not permitted for parts that are formed to a specific shape. Flat windows may be made from extruded sheet.

3.7.6.1.1.1 Optical uniformity. Windows shall be free from defects, such as imbedded particles, bubbles, scratches, blemishes, and striation, which will seriously affect the optical properties.

3.7.6.1.1.2 Corners. In the process of fabrication of windows, the three-sided corners shall not be reduced in excess of 30 percent of the nominal window thickness specified in the applicable specification sheet. This wider tolerance shall be limited to an area bounded by the hypotenuse of three right-angled triangles, ¾ by ¾ inch, placed so that their vertex is at the point of the outside surface of the corner.

3.7.6.1.2 Louver assembly. The louver assembly shall be as specified in the applicable specification sheet and herein. Finish shall be as specified in 3.8.4.2.

3.7.6.1.2.1 Material. Material for louver assemblies shall be aluminum. Mounting hardware and grounding straps are excluded.

3.7.6.1.2.2 Attachment to window. The louver assembly shall be attached to the window as specified on figure 4 for attaching louvers to watertight plastic windows.

3.7.6.1.2.3 Louver grounding. The louver assembly shall be grounded to the fixture housing through the mounting screws. Louver grounding shall be continuous from one mounting screw to the other. See MIL-DTL-917 for requirements for use of fasteners for continuity of current.

3.7.6.1.3 Window securing screw assemblies. Windows shall be provided with window-securing screw assemblies as specified on figures 1, 2, and 3. The type and number to be provided shall be as specified in the applicable specification sheet. When window assemblies are purchased separately for replacement parts, the following procedures shall be followed (this does not apply to windows shown in specifications sheets 18, 65, and 66 of MIL-DTL-16377):

a. The center securing screw assembly(ies) (all except those at the two extreme ends) shall be snugly fitted with a flat washer, a lock washer, and a nut as shown on figures 1 and 2.

b. A type VI label shall be provided with each window assembly as specified in 3.9.1.8.6.

3.7.6.1.4 Notches. When dimples or pins (as specified on figure 7), are used to retain window gasket in its groove, notches to clear these dimples or pins shall be filed on the sides of the window as shown in the applicable specification sheet and figure 27. Notches are not required when a continuous gasket retainer is used. Notches are required on windows when they are furnished separately from the lighting fixtures as replacement parts.

3.7.6.2 Windows and window assemblies for type III luminaires. Unless otherwise specified in the applicable specification sheet, windows and window assemblies for type III luminaires may be designed to the manufacturer's discretion. Unless otherwise specified in the applicable specification sheet, type III luminaires shall not include prismatic windows.

3.7.6.2.1 Material for windows. Material for windows shall be at the manufacturer's discretion, provided it meets the requirements of 3.4.

3.7.6.2.1.1 Optical uniformity. Windows shall be free from defects such as imbedded particles, bubbles, scratches, blemishes, and striation, which will seriously affect the optical properties.

3.7.6.2.2 Louver assembly. When a louver assembly is required for directional light output, it shall be as specified in the applicable specification sheet and herein. Finish shall be as specified in 3.8.4.2.

3.7.6.2.2.1 Material. Material for louver assemblies shall be aluminum. Mounting hardware and grounding straps are excluded.

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3.7.6.2.2.2 Attachment to window. The louver assembly shall be attached to the window as per the manufacturer's discretion. Attaching louvers to watertight luminaires shall not compromise the watertight integrity of the luminaire.

3.7.6.2.2.3 Louver grounding. The louver assembly shall be grounded to the fixture housing through the mounting devices. See MIL-DTL-917 for requirements for use of fasteners for continuity of current.

3.7.7 Lenses, globes, roundels, and windows for type II fixtures. Plastic lenses and windows shall be as specified in the applicable specification sheet. Glass lenses, globes, roundels, and windows (referred herein as glassware) shall be as specified in the applicable specification sheet and herein.

3.7.7.1 Optical uniformity. Glassware shall be free from defects that will prevent conformance with the light output requirements specified in the applicable specification sheet. Finished glassware shall be free from air bubbles, striae, wrinkles, mold marks, chipped edges, and any other blemishes which may affect the optical qualities (see 4.8.4).

3.7.7.2 Material. Glassware shall be made of high-impact, heat-resistant glass of a type that will withstand all requirements specified in the applicable specification sheet. The glass shall be of such composition and quality as to withstand exposure to atmospheric sea conditions (particularly that of salt water), high humidity, and bright sunlight, without etching, discoloration, or any change in the light transmittance. Flashed glass shall not be used.

3.7.7.3 Edges. The edges of all glassware shall be ground or molded to a true surface.

3.7.7.4 Colored glassware. Colored glassware shall be colored material throughout the entire thickness of the portions designed for light transmission.

3.7.7.5 Painted glassware. Those portions of the glassware designated by the applicable specification sheet to be painted shall be uniformly coated with inorganic baking enamel. The color shall be as specified in the applicable specification sheet. The glassware shall be baked at such a temperature and period of time as to ensure adhesion of the enamel to the glass. There shall be no evidence of loosening or softening or other dissolution of the enamel when glassware is tested as specified in 4.8.17.2.

3.7.7.6 Mechanical shock. Glassware shall be resistant to one impact of a freefalling 3-pound polished steel ball striking the glassware as specified in 4.8.17.3.

3.7.7.7 Thermal shock. Glassware shall withstand the stresses produced when tested as specified in 4.8.17.4, without visible or structural damage.

3.7.7.8 External hydrostatic pressure. Glassware shall withstand the stresses produced when tested as specified in 4.8.17.5, without visible or structural damage.

3.7.7.9 Breakage pattern. Breakage pattern shall be characterized by complete fracture of the glassware into small rectangular pieces predominantly not exceeding 2½ by ⅝ inches when tested as specified in 4.8.17.6.

3.7.8 Reflectors. Reflectors shall be as specified in the applicable specification sheet, or of the size and shape necessary to produce the distribution curve required (see 3.7.1 and 3.7.2). Reflectors for type I fixtures shall be made of aluminum sheet (the manufacturer's choice) not less than 0.019 inch thick. Reflectors shall be reinforced to prevent noise when a fixture is tested as specified in 4.8.9. Clearance of holes of a size to accept the socket wrench for ¼-20 hexagon nut shall be cut in the reflector where necessary to facilitate replacement of shock mounts in type I or type III, class 2 fixtures. The holes shall align with shock mount hexagon nuts when the reflector is assembled in the enclosure. For finish of reflectors, see 3.8.3. Material for specular reflectors shall be aluminum pre-finished lighting sheet having a specular reflectance factor of not less than 80 percent. Plastic coating for specula reflecting surface is not acceptable.

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3.7.9 Filters. Colored filters shall be as specified in the applicable specification sheet and herein.

3.7.9.1 Red filter tubes for type I, class 2 fixtures. When specified (see 6.2), or when required by the applicable specification sheet, red filter tubes for insertion over type I lamps for use in red illumination shall be furnished installed in the fixtures. The procedures specified herein shall apply when red filter tubes are installed in the type I standard fixtures.

3.7.9.1.1 All red fixture(s). The designation "(R)" shall be placed next to the symbol number specified in the applicable specification sheet to signify an all-red fixture; that is, red filter tubes have been installed on all lamps.

3.7.9.1.2 Single red with white fixture(s). The designation "(R&W)" shall be placed next to the symbol number specified in the applicable specification sheet to signify a combination red and white fixture; that is, a red filter tube has been installed on one lamp of a two-lamp fixture or on the center lamp of a three-lamp fixture.

3.7.9.1.3 Two red with white fixture(s). The designation "(2R&W)" shall be placed next to the symbol number specified in the applicable specification sheet to signify a combination red and white fixture; that is, red filter tubes have been installed on the two outer lamps of a three-lamp fixture.

3.7.9.1.4 Red directional fixture(s). Installation of red filter tubes in directional fixtures shall be as specified in the applicable specification sheet. The designation "(R&W)" shall be placed next to the symbol number specified in the applicable specification sheet to signify combination red and white fixture.

3.7.9.2 Blue filter tubes for type I fixtures. Identical requirements specified in 3.7.9.1 apply, except the designation "(B)" is substituted for red "(R)."

3.7.9.3 Gray filter tubes for type I fixtures. Identical requirements specified in 3.7.9.1 apply, except the designation "(G)" is substituted for red "(R)."

3.7.9.4 Night vision-friendly filter tubes for type I fixtures. Identical requirements specified in 3.7.9.1 apply, except the designation "(NV)" is substituted for red "(R)."

3.7.9.5 Amber filter tubes for type I fixtures. Amber filter tubes are used to replace red filters during peace time, as appropriate. There is no change to the symbol number of the fixture. The nomenclature of 3.7.9.1 applies.

3.7.10 Type III lens color. New design type III and legacy type II replacements luminaires may use a clear lens for red, amber, green, blue, or cyan lighting applications. The visible LED circuit board shall identify the illumination color.

3.7.11 Lumen maintenance for type III luminaires. Unless otherwise specified (see 6.2), luminaires shall be designed to have a minimum lumen maintenance of 70 percent of their initial output at 36,000 hours (see 6.9) in a 77 °F (25 °C) environment and 10,000 hours in a 122 °F (50 °C) environment when projected using the method described in IES TM-21 (with corresponding IES 28 data). Projections supported by IES TM-28 (with corresponding IES LM-84 data) will also be accepted with the same required minimum L_{70} projections. Measurements shall be conducted in accordance with 4.8.20.1 and 4.8.20.2 to support the lumen maintenance projections. The LEDs selected for use in a luminaire shall have available IES LM-80 test data for a minimum of 6,000 hours, with a minimum of 20 LED packages having been tested for the IES LM-80 report. The IES LM-80 test data shall contain case temperatures and drive currents that meet or exceed the luminaires measured in situ LED temperature measurement point (TMP_{LED}) (see 6.8.10) and drive current for compatibility with the lumen maintenance projection method of IES TM-21. The IES LM-80 report shall include one or more photographs or diagrams indicating the manufacturer's designated (TMP_{LED}) (see 6.2 and 6.4.1.1 for reporting of lumen maintenance information).

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3.7.11.1 Lumen maintenance, ambient (at 77 °F [25 °C]). When tested in accordance with 4.8.20.1, the following criteria shall be met:

a. The measured in situ TMP_{LED} temperature shall be equal to or less than the temperature specified in the IES LM-80 test report for the relevant drive current (see 4.8.20.1).

b. The drive current measured during the test shall be equal to or less than the drive current specified in the IES LM-80 test report for the relevant TMP_{LED} temperature.

c. Unless otherwise specified (see 6.2), the IES TM-21 lumen maintenance prediction shall project an L_{70} greater than 36,000 hours.

3.7.11.2 Lumen maintenance, accelerated (at 122 °F [50 °C]). The high-temperature testing shall be used to demonstrate that the luminaire is not adversely affected by the high-temperature environments occasionally found aboard ships. When tested in accordance with 4.8.20.2, the following criteria shall be met:

a. The measured LED in situ TMP_{LED} temperature shall be equal to or less than the maximum temperature specified in the IES LM-80 test report for the relevant drive current (see 4.8.20.2).

b. Unless otherwise specified (see 6.2), the IES TM-21 lumen maintenance prediction shall project an L_{70} greater than 10,000 hours.

3.8 Processes.

3.8.1 Treatment and processing of metals for corrosion resistance. The treatment and processing of metals for corrosion resistance shall be in accordance with MIL-DTL-917.

3.8.2 Painting. Painting of metals to obtain corrosion resistance shall be in accordance with MIL-DTL-917. See MIL-DTL-917 for parts to be painted or not painted.

3.8.3 Finishing for fixtures and parts. When the salt spray fog is specified in the applicable specification sheet, the procedures of 3.8.1 and 3.8.2 for finishing fixtures and parts for corrosion resistance shall be followed. When the salt spray fog is not specified in the applicable specification sheet, the procedures for finishing fixtures specified herein shall apply.

3.8.3.1 Cleaning. Cleaning shall be in accordance with the manufacturer's guidelines.

3.8.3.2 Surface preparation. The surface of metal parts shall be prepared for painting in accordance with the manufacturer's guidelines. This operation may be combined with the cleaning specified in 3.8.3.1.

3.8.3.3 Finish. Final finish shall be baking enamel of a commercial grade or powder coating in accordance with MIL-PRF-24712. Color shall be as specified in 3.8.3.4 through 3.8.3.6.

3.8.3.4 Types I, II, and type III, class 1 fixtures. Unless otherwise specified in the applicable specification sheet, the color of the final finish applied to all exterior surfaces of types I, II and type III, class 1 fixtures shall be haze gray in accordance with color number 26270 of SAE AMS-STD-595; the color of the final finish applied to all reflecting interior surfaces shall be white in accordance with color number 17875 of SAE AMS-STD-595, having a reflectance factor of not less than 80 percent. Specular reflectors shall not be painted (see 3.7.6).

3.8.3.5 Types I, II, and type III, class 2 fixtures. Unless otherwise specified in the applicable specification sheet, external and reflecting surfaces of types I, II, and type III, class 2 fixtures shall be white in accordance with color number 17875 or 17925 of SAE AMS-STD-595, having a reflectance factor of not less than 80 percent. Specular reflectors shall not be painted (see 3.7.8).

3.8.3.6 Parts. Unless otherwise specified in the applicable specification sheet, parts shall be painted light gray in accordance with color number 16492 of SAE AMS-STD-595.

3.8.4 Finishing for windows and window assemblies for type I fixtures. Finishing for windows and window assemblies for type I fixtures shall be as specified in the applicable specification sheet and herein.

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3.8.4.1 Windows. Surfaces of areas specified in the applicable specification sheet to be finished shall be finished with air dry acrylic lacquer or an air dry water-based paint as follows:

a. White finish: Highly diffusing white. Luminous transmittance of the finished surface shall be 60±10 percent.

b. Black finish: Flat black. Surfaces shall be completely covered so that no pinholes or other blemishes exist.

c. Frosted finish: Clear frosted. Luminous transmittance of the finished surface shall be 65±10 percent.

3.8.4.2 Louver assemblies. After surface preparation (see 3.8.3.2), the final finish for louver assemblies shall be baking enamel of a commercial grade or powder coat in accordance with MIL-PRF-24712. The color shall be as follows:

a. White louvers: White in accordance with color number 17875 of SAE AMS-STD-595, having a reflectance factor of not less than 80 percent.

b. Black louvers: Dull camera black.

3.8.4.3 Antistatic coating. Antistatic coating shall be applied to all unfinished surfaces of the window assemblies.

3.8.5 Stress relief.

3.8.5.1 Stress-corrosion cracking of metals. Stress-corrosion cracking characteristics are of primary concern in material selection for marine service. High residual stresses in tension in certain materials can cause stress-corrosion cracking when exposed to corrosive environment. Stress-corrosion cracking occurs under tensile stresses which are induced into metal parts that are formed by bending or drawing or that are fabricated by welding or torch welding. All susceptible metal parts under tensile stress shall be stress relieved to prevent deterioration or failure. Method of stress relieving shall be the manufacturer's choice as required by the technical data furnished by the supplier of the raw material.

3.8.5.2 Plastics. Measures shall be taken in molding or processing plastics to ensure that stress buildup does not occur or is satisfactorily treated to relieve these stresses to prevent deterioration or failure of a part or assembly. The stress-relieving process shall be as required by the technical data furnished by the manufacturer of the raw material.

3.8.5.3 Plastic windows for type I and type III fixtures. Plastic windows for type I and type III fixtures that are formed to a specified shape from sheet shall be annealed to relieve internal stresses after fabrication and finishing operations have been completed. After annealing, there shall be very little or no evidence of stress when window assemblies are tested as specified in 4.8.5. Very little stress is defined as that which does not interfere with the optical uniformity of the window and which does not cause window cracking under the shock and vibration tests. Method of annealing shall be the manufacturer's choice as prescribed by the technical data furnished by the manufacturer of the raw material.

3.8.6 Welding. Welding and allied processes used in fabrication of fixtures and parts shall be based on the American Welding Society Handbook as a guide.

3.9 Designation and marking. Labels, identification, information and warning plates, instruction sheets, tags, and markings for fixtures and parts shall be as specified in the applicable specification sheet and herein. Labels and identification plates are not required with plastic fixtures or parts where the specified information is molded in place.

3.9.1 Labels. Labels shall be furnished as specified herein. Where advantageous, labels may be combined.

3.9.1.1 Material. The labels shall be of a grade of paper, fabric, foil, or plastic, flatback, coated on one side with pressure-sensitive adhesive. The texture of material shall permit flexibility and shall have a finish capable of withstanding normal usage of handling and installation.

3.9.1.2 Adhesive. The adhesive shall be a pressure-sensitive, permanent type. It shall be water-insoluble, homogeneous, and shall be coated in a smooth layer on one side of the label. The adhesive shall require no solvent, heat, or other preparation prior to application. Adhesive shall adhere to metal or plastic.

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3.9.1.2.1 Adhesion. The label shall adhere to the clean, dry surfaces to which applied under normal pressure, without curling, breaking, or lifting.

3.9.1.3 Color. Unless otherwise specified herein, the color of the labels shall be white, yellow, or silver.

3.9.1.4 Thickness. Thickness of material prior to coating with adhesive shall be sufficient to give structural body for printing and handling.

3.9.1.5 Size. The size of labels shall accommodate the desired information.

3.9.1.6 Writing quality. Labels shall allow printing or writing with ink.

3.9.1.7 Marking. Marking shall be with black waterproof ink. Other colors of ink may be used when contrast is desired. Non-waterproof ink may be used, provided it is followed by the application of a transparent waterproof coating. Height of characters shall be approximately as shown on figure 11 for type I labels and not less than 1/16 inch for types II, III, IV, and VI labels.

3.9.1.8 Application and classification. Unless otherwise specified in the applicable specification sheet, labels shall be applied as specified herein. Labels shall be of the types specified herein and as specified in the applicable specification sheet.

3.9.1.8.1 Type I labels. Type I labels shall be for equipment identification and shall be as shown on figure 11. Unless otherwise specified in the applicable specification sheet, two type I labels shall be applied to each fixture; one located on the outside back surface and the other located on an inner nonreflecting surface.

3.9.1.8.2 Type II labels. Type II labels for use with types I and III fixtures that are provided with three (or more) conductor cables for either single white or red or combination white and red illumination shall be as shown on figure 12. Type II labels shall be applied on the outside surface adjacent to the cable entrance of each flush-mounted fixture.

3.9.1.8.3 Type III labels. Unless otherwise specified in the applicable specification sheet, type III labels shall be used to indicate internal wiring of fixtures and shall be as shown on figure 13. Type III labels shall be applied as specified in the applicable specification sheet.

3.9.1.8.4 Type IV labels. Type IV labels shall be used for supplying information pertinent to the fixtures or parts and shall be as shown on figure 14. The information to be imprinted thereon and the application to fixtures or parts shall be as specified in the applicable specification sheet.

3.9.1.8.5 Type V labels. Type V labels shall be as shown on figure 15 and shall be used for warning information. The warning information and the application to fixtures or parts shall be as specified in the applicable specification sheet.

3.9.1.8.6 Type VI labels. Type VI labels shall be as shown on figure 16. Unless otherwise specified in the applicable specification sheet, one type VI label shall be applied on the outside of each window assembly that is furnished as a replacement part for type I fixtures. The adhesive for these labels shall be of a pressure-sensitive type that will allow the label to be easily removed (this does not apply to windows shown in specification sheets /18, /65, or /66 of MIL-DTL-16377).

3.9.2 Identification and information plates. When specified in the applicable specification sheet, identification and information plates shall be furnished in accordance with MIL-DTL-15024 and MIL-P-15024/5. Identification plates for use on fixtures that will be exposed to the weather shall be type A, B, C, D, or I, made from nickel-copper alloy, brass, or type H made from anodized aluminum. Types E and G plates shall be used only for wiring diagrams and other plates that are housed within the equipment enclosure. Plastic type B plates shall have black inner (or record) laminations and gray cover laminations.

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3.9.3 Identification plate (explosionproof). An identification plate in accordance with MIL-DTL-15024 and MIL-P-15024/5, bearing the following legend, in upper case letters, shall be provided for fixtures and lights having explosionproof or combined explosionproof-watertight enclosures:

"FOR USE IN THE FOLLOWING EXPLOSIVE
ATMOSPHERES: GASOLINE, PETROLEUM,
NAPHTHA, ALCOHOL, ACETONE, LACQUER
SOLVENT VAPORS, AND NATURAL GAS."

The identification plate shall be securely attached to the outside of the fixture where it can best be seen. Where space permits, this legend may be combined with the aforementioned identification and information plate (see 3.9.2).

3.9.4 Instruction sheets or tags. When specified in the applicable specification sheet, instruction sheets or tags providing pertinent design and installation information shall be furnished with the fixtures or parts.

3.10 Interchangeability and standardization.

3.10.1 Interchangeability. Each repair part shall be equal to the original and shall allow replacement without special tools (see 6.8.7) and without the removal of rivets or pins, dependent on a force fit. Mechanical and electrical interchangeability shall exist between like assemblies, subassemblies, units, and replaceable parts, regardless of the manufacturer. Interchangeability does not mean the original and replacement parts are identical; interchangeability requires that a substitution of such like assemblies, subassemblies, units, and replaceable parts be easily effected without physical or electrical modifications to any parts of the lights or assemblies, including globes, lenses, roundels, lamps, lampholders, wiring, and mounting and without resorting to selection.

3.10.2 Standardization.

3.10.2.1 Component standardization. Standardization of units, assemblies, subassemblies, and parts shall be accomplished to as great an extent as practicable to simplify supply parts and replacement problems. Electrical and mechanical interchangeability of units, assemblies, subassemblies, and parts shall conform in detail to the various individual specification sheets.

3.10.2.2 Equipment standardization. Progressive standardization of complete equipment is required, at least to the extent that similar articles of different manufacture are electrically and mechanically interchangeable. To accomplish maximum practicable standardization, individual specification sheets will include requirements as to mounting dimensions, limiting overall dimensions, and electrical characteristics of individual items of equipment. Mounting dimensions shall be as specified in the applicable specification sheet and figures 25, 26, and 28.

3.10.2.3 Standard stock parts. Standard stock parts (see 6.8.8) and hardware shall be used to the greatest extent practicable.

3.10.2.4 Proprietary parts. Where Navy standard stock parts are not suited for the purpose intended, parts and materials shall be used which are standard, easily obtainable, and produced by reliable manufacturers, as far as possible within the limits of this specification. In order to facilitate the acquisition of replacement parts, the design shall not be based upon the use of parts of non-standard manufacture where units of standard manufacture are available. Similarly, the design shall not be based on the use of parts produced by only one manufacturer when an equivalent design available from several sources of manufacture might be employed.

3.11 High-temperature fluorescent lighting fixtures. When specified (see 6.2), the procedures specified herein shall apply when high-temperature fluorescent lighting fixtures are required.

3.11.1 Symbol numbers. The designation "(HT)" shall be placed next to the symbol number specified in the applicable specification sheets to indicate a high temperature modification of the standard lighting fixture (for example: Symbol Number 77.4HT).

3.11.2 Component substitution. Standard components specified in the applicable specification sheets shall be substituted with corresponding high-temperature components listed in table III. Other features, except National Stock Numbers, shall remain as specified in the applicable specification sheets.

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TABLE III. Component substitution for high-temperature fluorescent lighting fixtures.

Component	Fluorescent Lighting Fixtures Using the 6 or 8 Watt Fluorescent Lamps		Fluorescent Lighting Fixtures Using the 15 or 20 Watt Fluorescent Lamps	
	Use	In Lieu of	Use	In Lieu of
Lampholders	UL 496 (naval-use lampholders, 302 °F [150 °C])	UL 496 (naval-use lampholders, 221 °F [105 °C])	UL 496 (naval-use lampholders, 302 °F [150 °C])	UL 496 (naval-use lampholders, 221 °F [105 °C])
Starter sockets	UL 496 (naval-use starter holders, 302 °F [150 °C])	UL 496 (naval-use starter holders, 221 °F [105 °C])	UL 496 (naval-use starter holders, 302 °F [150 °C])	UL 496 (naval-use starter holders 221 °F [105 °C])
Ballasts	M16377/45-002	M16377/45-001	M16377/72-002	M16377/72-001
Internal wiring	Wire with silicone impregnated fiber-glass insulation (SF-2, 302 °F [150 °C] wire)	Wire with thermoplastic insulation	Wire with silicone impregnated fiber-glass insulation (SF-2, 302 °F [150 °C] wire)	Wire with thermoplastic insulation
Wire connectors	Connectors with polyvinylidene fluoride (PVDF)	Connectors with nylon insulation	Connectors with PVDF	Connectors with nylon insulation
Supply cable	Type LS7PS-6 of MIL-DTL-24643/26	Type LS7SGU-3 of MIL-DTL-24643/20	Type LS7PS-6 or of MIL-DTL-24643/26	Type LS7SGU-3 of MIL-DTL-24643/20
Cable entrances	Aluminum washers and cap. Dimensions shall be as specified in ASTM F1836M.	Nylon washers and cap as specified in ASTM F1836M.	Aluminum washers and cap. Dimensions shall be as specified in ASTM F1836M.	Nylon washers and cap as specified in ASTM F1836M.

3.12 Workmanship. The fixtures, lights, and parts shall be manufactured and finished in a manner that is consistent with best industry practices. Particular attention shall be paid to neatness and thoroughness of soldering, crimping, wiring, marking of parts and assemblies, plating, painting, machine screw assemblage, welding, and freedom of parts from burrs and sharp edges. See the workmanship guideline of MIL-HDBK-454 for additional information.

3.13 Energy efficiency. The Navy's energy conservation program encourages the design of lightweight and energy-efficient luminaires. Energy efficiency for new luminaire development shall be based on acquisition requirements (see 6.2) and applicable specification sheets. The following energy efficiency requirement is only applicable to applications using type III luminaires as replacements for types I and II luminaires. Electrical energy usage for replacement type III luminaires shall be measured in watts and, for fixtures with power greater than 10 watts, shall not consume more than 50 percent of the energy of the replaced legacy luminaire.

3.14 Predicted lifetime for drivers for type III luminaires. Drivers (see 6.8.1) for type III luminaires shall be designed for a mean time between failures (MTBF) of 50,000 hours in a naval-sheltered environment at 77 °F (25 °C). MIL-HDBK-217 may be used as guidance for calculation of MTBF.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Comparison inspection (see 4.3.4).

c. Conformance inspection (see 4.4).

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4.2 Inspection conditions. Unless otherwise specified (see 6.4), all inspections shall be performed in accordance with the test conditions specified in 4.8 and the applicable specification sheets.

4.3 First article inspection. First article inspection shall consist of the examination of 4.5 and the tests of 4.8 and as specified in the applicable specification sheets.

4.3.1 Order of tests. For watertight and submersible enclosures, the following tests shall be performed in the order specified:

a. Dielectric withstanding voltage (see 4.8.2).
b. Insulation resistance (see 4.8.3).
c. Enclosure effectiveness (see 4.8.14).
d. Shock (see 4.8.7).
e. Vibration (see 4.8.8).
f. Enclosure effectiveness (see 4.8.14).
g. Dielectric withstanding voltage (see 4.8.2).
h. Insulation resistance (see 4.8.3).
i. Noise (when applicable) (see 4.8.9).

For enclosures other than watertight and submersible, the tests shall be performed in the same order, except that the tests for dielectric withstanding voltage, insulation resistance, and effectiveness of enclosure are not required prior to tests for shock and vibration.

4.3.2 Inspection of parts. When shock, vibration, and effectiveness of enclosure tests are specified in the applicable specification sheets for parts (window assemblies and glassware) that are an integral part of a fixture or light and are furnished as replacement parts, these parts shall be installed in a fixture or light for the performance of these tests.

4.3.3 Combination of tests. Where advantageous, tests may be combined in cases where the basic body of the fixture or light to be tested is the same for different styles of the fixture or light. For example, in the case where a red globe is substituted for a diffusing globe to obtain red illumination in the incandescent fixtures or a clear white shielded window is substituted for a diffusing one in the fluorescent fixtures.

4.3.4 Comparison inspection. One sample from the first production run on each fixture, light, or part shall be subjected to the same inspection specified in the applicable specification sheets for first article inspection when specified (see 6.2 and 6.5). Procedures for first article inspection specified in 4.3 shall also apply for comparison inspection.

4.4 Conformance inspection. Conformance inspection shall consist of the examination and tests of 4.5 and 4.6.

4.4.1 Sampling. Sampling shall be as specified in 4.4.1.1 and 4.4.1.2.

4.4.1.1 Lot. For the purpose of sampling, a lot shall be the identical number of items on the contract or order.

4.4.1.2 Sampling for examination and conformance tests. As a minimum, a sample quantity of fixtures shall be randomly selected from each lot in accordance with table IV and examined as specified in 4.5 and 4.6. Detection of any nonconforming characteristic in any sample shall result in the rejection of the entire lot. As an option, 100 percent of the lot may be screened for the defective characteristic(s) or a new lot may be provided, which shall be inspected in accordance with the sampling plan contained herein.

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TABLE IV. Sampling for examination and conformance tests.

Lot Size	Sample Size
2 to 4	1
5 to 50	5
51 to 90	7
91 to 150	11
151 to 280	13
281 to 500	16
501 to 1,200	19
1,201 to 3,200	23
3,201 to 10,000	29
10,001 to 35,000	35
35,001 and over	40

4.5 Examination. Fixtures, lights, or parts shall be examined as specified in 4.5.1 and 4.5.2.

4.5.1 Visual examination. Fixtures, lights, or parts shall be examined for the defects listed in table V, as applicable.

TABLE V. Visual defects.

Examine	Defect
Finish	Finish not as specified: blistered, peeled, chipped, area of no film or insufficient diffusion. Color not as specified.
Materials	Materials not as specified. In addition, prohibited materials are used (see 3.4.2 and 3.4.2.1).
Construction and workmanship, general (applicable to all parts and assemblies)	Part missing, fractured, split, punctured, dented, or deteriorated. Not in proper alignment. Sharp burr or edge, sliver, or splinter.
Welding and brazing (where applicable)	Missing, incomplete, burn holes, cracked, fractured, or otherwise not fused, slag inclusion, slight undercut, not smooth and uniform, scale or flux deposits not removed.
Soldering (when applicable)	Missing, not adherent or incomplete. Not clean (flux or flux residue not removed); not smooth (surface not finished neatly), or pinholes in solder.
Bolts, nuts, screws, studs, pads, rivets, and other types of fasteners	Missing, broken, stripped, fractured, loose, bent, not peened or insufficiently peened.
Electrical assembly	Connections at screw terminals and splice connection not as specified. Wiring not properly joined, loose at terminal, or not enclosed in specified conduit where required. Adequate slack not provided for wiring to relieve similar or excessive insulation stripped from wiring.
Gaskets (where applicable)	Not coated where required, missing, not as specified.
Marking for identification	Missing, incomplete, not legible.
Tag or sheet (where applicable)	Missing, incomplete, not legible.

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4.5.2 Examination for weight and dimensions. Fixtures, lights, or parts shall be examined to determine compliance with weight and dimensional requirements. Weights or dimensions not within specified tolerances shall be classified a defect.

4.6 Conformance tests. Fixtures, lights, or parts shall be subjected to the tests specified in 4.6.1 through 4.6.3.

4.6.1 Complete fixtures or lights. Sample fixtures or lights shall be subjected to the operation, dielectric withstanding voltage, and insulation resistance tests of 4.8.1, 4.8.2, and 4.8.3. Additionally, type III luminaires shall be subjected to the chromaticity test of 4.8.6.5.

4.6.2 Plastic windows for type I fixtures and lights. Sample plastic windows for type I fixtures and lights shall be subjected to the optical uniformity and stress relief tests of 4.8.4 and 4.8.5. These tests shall be performed when the windows are furnished as an integral part of a fixture or light or as a repair part.

4.6.3 Lenses, globes, and roundels for type II fixtures and lights. Lenses, globes, and roundels for type II fixtures and lights shall be subjected to the optical uniformity test of 4.8.4. This test shall be performed when the lenses, globes, and roundels are furnished as an integral part of a fixture or light or as repair parts.

4.7 Packaging. See section 5.

4.8 Test procedures.

4.8.1 Operation. The fixtures, lights, and parts shall be electrically and mechanically operated to determine conformance to the requirements of this specification.

4.8.2 Dielectric withstanding voltage. Fixtures, lights, or parts shall be subjected for 1 minute to a dielectric test voltage, the effective potential of which is twice rated root mean square (RMS) voltage plus 1,000 volts, applied between points of polarity and also between live parts and ground. The frequency of the test voltage shall be 60 hertz alternating current and shall approximate a true sine wave. (A test potential of twice rated voltage plus 500 volts RMS may be used for fixtures or lights rated 28 volts or less.)

4.8.3 Insulation resistance. After the dielectric withstanding voltage test (see 4.8.2), insulation resistance of the sample fixtures, lights, or parts shall be measured between all current carrying parts and non-current carrying parts with a megohmmeter at a voltage not less than 500 volts direct current, at normal room temperature. This test is not required for unwired fixtures wherein plastic parts approved by the Technical Authority are certified by the manufacturer as meeting the requirements of 3.6.16.

4.8.4 Optical uniformity test for windows, lenses, globes, and roundels. Windows, lenses, globes, and roundels furnished either as a part of a fixture or light or as replacement parts shall be examined to ascertain that the product is of uniform quality as to color, clarity, and freedom from dead zones or streaks. Unless otherwise specified (see 6.2), the source of light shall be a standard 100-watt, 120-volt, tungsten filament lamp operating at approximately normal rated voltage. This test may be combined with the operation test of 4.8.1 where the source of light shall be that of the fixture or light (see 3.7.7.1).

4.8.5 Stress relief test for plastic window for type I and type III fixtures and lights. The windows shall be placed between a pair of polarizing lenses and a light source to determine conformance to 3.8.5.3.

4.8.6 Light output. Unless otherwise specified (see 6.2), the test procedure for measuring light output shall conform to the appropriate IES guides for photometric and colorimetric testing given below and the applicable specification sheets. Type I luminaire testing shall be as directed by IES LM-41. Type II luminaire testing shall be as directed by IES LM-46. Type III luminaires shall be tested to IES LM-79.

4.8.6.1 Candlepower distribution. When specified in the applicable specification sheet, a candlepower distribution curve shall be obtained from the sample fixture or light. Before any measurements are taken, the fixture or light shall be operated until the light output has reached a stable condition.

4.8.6.2 Footcandle distribution. When specified in the applicable specification sheet, a footcandle distribution curve shall be obtained from the sample fixture or light. Before any measurements are taken, the fixture or light shall be operated until the light output has reached a stable condition.

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4.8.6.3 Color and luminous transmittance. When specified in the applicable specification sheet, the color and luminous transmittance of windows, globes, roundels, lenses, and filters shall be checked to determine conformance to 3.7.3.

4.8.6.4 Brightness test for type III luminaires. Brightness (luminance intensity) testing and reports shall be based on IES LM-79. Maximum brightness shall be determined using a Konica Minolta LS100/110 luminance meter, or equal, at the vertical angles 0, 45, 55, 65, 75, 85 and the lateral angles 0, 45, and 90 degrees.

4.8.6.5 Chromaticity test for type III luminaires. Chromaticity shall be determined in accordance with IES LM-79 (see 3.7.4).

4.8.6.6 CCT test for type III luminaires. CCT shall be determined in accordance with IES LM-79.

4.8.6.7 Color rendition test for type III luminaires. The color rendition shall be determined in accordance with IES TM-30.

4.8.6.8 Lumen output for type III emergency luminaires. The following procedure shall be used to measure the lumen output of type III emergency luminaires:

a. Ensure battery is at full charge by applying power for a minimum of 24 hours.
b. Remove external power source.
c. Take initial lumen measurement in accordance with IES LM-79.
d. Allow fixture to run on battery for 5 hours.
e. Take lumen measurement in accordance with IES LM-79.

4.8.7 Shock test. The sample fixture, light, or part shall be subjected to the type A, H.I. shock test in accordance with MIL-DTL-901 for grades A or B (see 3.5.12), class I for fixtures and lights without shock mounts, or class II for fixtures and lights with shock mounts, lightweight equipment. Fixtures or lights shall be energized during this test. Photographs shall be taken before the test of the test setup and after the test of any failed element.

4.8.7.1 Mounting. Mounting of fixtures, lights, and parts on the shock machine shall simulate normal shipboard installation. Mounting dimensions and hardware shall be as specified in the applicable specification sheet. Fixtures, lights, and parts shall be securely mounted on the shock machine in accordance with the standard methods shown on figure 17.

4.8.8 Vibration test. The sample fixture, light, or part shall be subjected to a vibration test in accordance with type I vibration of MIL-STD-167-1 (see 3.5.13). Mounting for fixtures and lights shall simulate shipboard installation. Fixtures or lights shall be energized during this test. Photographs shall be taken in accordance with MIL-STD-167-1.

4.8.9 Noise test. When required (see 3.5.14), after the vibration and shock tests are conducted, the sample fixture, light, or part shall be submitted to an airborne noise test as follows: Using a sound level meter that meets the requirements of ANSI/ASA S1.4, determine the ambient noise level 12 inches from the fixture, light, or part under test. The maximum acceptable ambient noise level is 40 dB. With the fixture, light, or part energized, determine the maximum combined (ambient plus test piece) noise level at a distance of 12 inches from the test piece in all directions. The combined noise shall fall within the shaded area of figure 20. The fixture, light, or part shall be mounted as in actual service during this test. The shipboard equipment noise section of MIL-STD-1474 may be used as a guide. Calibration of sound level meter and recording of results shall be in accordance with the shipboard equipment noise section of MIL-STD-1474.

4.8.10 Salt fog test. The sample fixture, light, or part shall be subjected to a salt fog test in accordance with method 509.7 of MIL-STD-810 (see 3.5.15). The salt solution concentration shall be 20 percent.

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4.8.11 Grounding circuit. The grounding circuit of the fixture, light, or part shall be tested by placing a current of not greater than 5 nor less than 4 amperes through the grounding circuit for a period of not less than 1 minute. This shall be done with the fixture, light, or part insulated from the ground conductor. The ground conductor of a 2-foot cable shall be connected to one end of the circuit, and the exposed metal portion of the fixture, light, or part shall be connected by means of a test clip to the other end of the circuit. This circuit shall be placed in series with an ammeter and some means of limiting the current to the specified value. The potential drop through the ground connection when carrying between 4 and 5 amperes shall be measured by means of a voltmeter having a full scale deflection of not more than 3 volts connected between a metal portion of the fixture, light, or part in electrical contact with the grounding circuit and the ground conductor of the cable. This potential shall be not greater than 0.25 volt, and no arcing or burning in the grounding circuit shall be evident. The resistance of the grounding circuit shall be measured with a milliohmmeter and the resistance of the circuit shall be not greater than 0.1 ohm.

4.8.12 Continuity of grounding. The resistance shall be measured with a milliohmmeter between the protruding tip of the shock mount and any remote internal metallic structure of the housing assembly of the fixture, light, or part.

4.8.13 Leakage current (portable fixtures only). The leakage current test shall be conducted with the fixture, light, or part operating at approximately the normal operating temperature. The metal portion of the fixture, light, or part shall be ungrounded. Prior to the leakage current test, the resistance from the grounding conductor of the extension cable to the exposed metal portion of the light shall be measured and found to be less than 0.1 ohm. For measurement of leakage current, all connections shall be made at the end of a 2-foot long cable. An adapter may be used to facilitate making connections to the power source and meter. To operate the fixture, light, or part, the power source shall be connected to the proper conductors of the cable; however, the grounding conductor shall be left unconnected except as required for this test. For leakage current measurements, each line shall be connected successively to the grounding contact of the cable and the current flowing in the connecting wire measured.

4.8.14 Enclosure effectiveness. Effectiveness of the enclosure of the sample fixture, light, or part shall be determined as specified herein (see 3.5.11).

4.8.14.1 Totally enclosed. No special tests are required. It shall be ascertained by examination that the enclosure effectively performs its function and that the openings are not larger than those permitted by MIL-STD-108.

4.8.14.2 Dripproof. Dripproof enclosures shall be tested in accordance with MIL-STD-108. The fixture, light, or part shall be mounted in normal position when tested. There shall be no leakage except as permitted by MIL-STD-108.

4.8.14.3 Splashproof. Splashproof enclosures shall be tested in accordance with MIL-STD-108. The fixture, light, or part shall be mounted in normal position when tested. There shall be no entry of water into the enclosure when the fixture, fight, or part is subjected to a stream of water as specified in MIL-STD-108.

4.8.14.4 Watertight and submersible. Watertight and submersible enclosures shall be tested in accordance with MIL-STD-108. Before conducting this test, joints that have gaskets or sealing compound, or that have been painted, and which are subjected to being opened and closed shall be opened and closed three times (this includes doors and covers but does not include indicator and observation windows). There shall be no leakage of water into the enclosure after it has been subjected to the test conditions of MIL-STD-108.

4.8.14.5 Explosionproof. Explosionproof enclosures shall be tested as specified in 4.8.15.

4.8.14.6 Combined explosionproof-watertight. Combined explosionproof-watertight enclosures shall be tested as specified in 4.8.14.4 and 4.8.15.

4.8.15 Explosionproof tests. After all tests have been conducted, sample fixtures and lights having explosionproof or combined explosionproof-watertight enclosures shall be subject to the following test.

4.8.15.1 Explosion test. A minimum of twenty explosion tests shall be performed in accordance with UL 844 for the appropriate hazardous location type (see 6.2). Rejections shall be based on the criteria in UL 844 for the intended environment.

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4.8.15.2 Hydrostatic pressure test. The test fixture shall pass the internal hydrostatic pressure test of UL 844.

4.8.16 Electromagnetic interference (types I and III only). Types I and III fixtures, lights, or parts shall be subjected to an electromagnetic interference test in accordance with MIL-STD-461.

4.8.16.1 Electromagnetic interference for class 2 emergency fixtures. Types I and III, class 2 emergency fixtures shall be subjected to the electromagnetic interference test in accordance with MIL-STD-461 at full light output with battery charging enabled.

4.8.17 Tests for glassware. Tests for glassware shall be conducted as specified herein.

4.8.17.1 Color determination test. One sample of each color of colored glassware shall be broken to determine conformance to 3.7.7.4.

4.8.17.2 Paint removal test. Sample glassware shall be submerged in any commercially available paint and varnish remover for a period of not less than 3 minutes to determine conformance to 3.7.7.5.

4.8.17.3 Mechanical shock test. Sample glassware shall rest with its flat surface on a soft 2-foot by 2-foot by 2-inch-thick wood plank as shown on figure 18. A 3-pound polished steel ball shall be allowed to fall free and strike the glassware on the center of the outside surface. Striking force shall be as specified in the applicable specification sheet (see 3.7.7.6).

4.8.17.4 Thermal shock test. Sample glassware shall be uniformly heated to a temperature of 212±3.6 °F (100±2 °C) for 1 hour. The samples shall then be immediately immersed completely in water at a temperature of 32 to 50 °F (0 to 10 °C). At least ten consecutive cycles shall be performed on each sample (see 3.7.7.7).

4.8.17.5 External hydrostatic pressure test. Sample glassware shall be installed on lights for which they are designed and shall then be submerged in water in a pressure tank. An external hydrostatic pressure as specified in the applicable specification sheet shall be applied for a period of 1 hour (see 3.7.7.8).

4.8.17.6 Breakage pattern test. Sample glassware shall rest with its flat surface on a soft 2-foot by 2-foot by 2-inch-thick wood plank as shown on figure 18. A steel ball shall be allowed to fall free and strike the glassware on the center of the outside surface with a force great enough to break the glassware (see 3.7.7.9).

4.8.18 Magnetic permeability. The magnetic permeability of the material shall be determined in accordance with ASTM A342/A342M (see 3.4.8).

4.8.19 Ambient temperature. The lights shall be operated for 100 hours at the ambient temperature and voltage specified in the applicable specification sheet (see 3.5.17). The light shall be fully assembled (with all gaskets, if applicable, in place).

4.8.20 Lumen maintenance measurements for type III luminaires. Measurements shall be accomplished using the following procedures.

4.8.20.1 Lumen maintenance, ambient (77 °F [25 °C]) measurement. Lumen maintenance measurements shall be accomplished using the following procedure and data from the LED manufacturer:

a. The luminaire shall be mounted in its proposed orientation in a 77 °F (25 °C) ambient environment (see 6.2).

b. The ambient temperature shall be stabilized at 77 °F (25 °C) and maintained within ±3.6 °F (±2 °C) throughout the measurement.

c. The hottest LED device in the luminaire shall be determined based on the recommendations contained in IES LM-84.

d. A thermocouple shall be attached to the hottest LED device determined in step c. at the measurement point indicated in the IES LM-80 report (TMP_{LED}) provided by the LED manufacturer. Access may be through a minimally sized hole in the housing/array assembly, resealed with a sealant. Photographs of thermocouple placement and location of the hottest LED device in the luminaire shall be taken.

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e. The luminaire shall be turned on and the temperature allowed to stabilize. Stabilization is achieved after a minimum of 7.5 hours of runtime or after a minimum of 3 hours if three successive readings of TMP_{LED} taken at 15-minute intervals are within 1.8 °F (1 °C) of each other and not rising.

f. When stabilization is achieved, the following shall be measured at the designated LED:

(1) Voltage drop (V) across the LED

(2) Current (I) in series with the LED

(3) TMP_{LED}

g. The values obtained in step f. shall be entered into the IES TM-21 calculator provided by the Department of Energy for lumen maintenance prediction (see 3.7.11).

4.8.20.2 Lumen maintenance, accelerated (122 °F [50 °C]) measurement. Measurements shall be taken as specified in 4.8.20.1, with the ambient temperature increased to 122 °F (50 °C).

4.8.21 Power interface test. The fixture shall be subjected to power interface testing in accordance with MIL-STD-1399-300-1 (see 3.6.18).

4.8.22 Fail-safe circuit design test. One LED shall be selected at random within the luminaire; open circuit the LED, and verify all remaining LEDs remain operational and luminaire light output is not visibly degraded.

4.8.23 Class 2 emergency light fixture battery and charging system testing. The following procedure shall be used to test the charging capability of an emergency light fixture. Testing shall be accomplished at an ambient temperature of 131 °F (55 °C).

a. The battery shall be charged to its maximum charge over a period of 24 hours via 115 volts alternating current (VAC), 60 hertz being applied to the fixture wiring. The initial voltage of the battery pack shall be measured and recorded as V_i.

b. The input power to the fixture shall be removed. The fixture shall now be running on battery power. The battery voltage and ambient temperature shall be recorded every 30 minutes.

c. The fixture shall be run on battery power for 5 hours. The battery voltage, when measured after 5 hours (V_{B1}), shall be not less than 87.5 percent of its initial voltage (V_i).

d. 115 VAC, 60 hertz input power shall be reapplied to the fixture for 24 hours to recharge the battery.

e. The input power shall be removed and the fixture shall be run on battery power for a period of 3 hours and 20 minutes. The battery voltage, when measured after 3 hours and 20 minutes (V_{B2}), shall be not less than 87.5 percent of its initial voltage (V_i). The battery voltage and ambient temperature shall be recorded every 30 minutes.

f. 115 VAC, 60 hertz input power shall be reapplied to the fixture for 24 hours to recharge the battery.

g. The input power shall be removed. The fixture shall be run on battery power until the low battery voltage cutoff engages. The battery voltage and ambient temperature shall be recorded every 30 minutes.

h. 115 VAC, 60 hertz input power shall be reapplied to the fixture for 24 hours to recharge the battery.

i. The input power shall be removed and the fixture shall run on battery power for 5 hours. The battery voltage, when measured after 5 hours, shall be not less than 87.5 percent of its initial voltage (V_i). The battery voltage and ambient temperature shall be recorded every 30 minutes.

4.8.24 Prohibited materials. Prohibited materials shall be verified as specified (see 3.4.2 and 6.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Fixtures designed and manufactured to this specification and the associated specification sheets are intended for use in a naval shipboard environment, both interior and exterior of the ship's structure, to provide light during both normal and emergency conditions. The fixtures are able to perform in high temperatures, after shock events, under constant vibration, and in casualty situations. The extensive electrical performance testing ensures the electromagnetic radiation of current and future fixture designs will not adversely interfere with the advanced communications and combat systems employed on multiple platforms.

6.1.1 Class 1 fixtures. Class 1 fixtures and lights are intended for use in detail lighting aboard ship. Examples of such applications, among others, are mirror and berth illumination, desk lighting, gaugeboard, and spot illumination.

6.1.2 Class 2 fixtures. Class 2 fixtures and lights are intended for use in general illumination aboard ship. Fixtures and lights for general white illumination are also used for red or blue illumination by the use of proper filters inserted in the lighting fixtures.

6.1.3 Overhead standard-mounted fixtures. Overhead standard-mounted fixtures and lights are intended to be mounted on the overhead by standard shock mounts.

6.1.4 Overhead flush-mounted fixtures. Overhead flush-mounted fixtures and lights are intended to be flush-mounted in areas where sheathing is employed in the overhead.

6.1.5 Explosionproof fixtures. Explosionproof fixtures and lights are intended for use in areas where explosive atmospheres exist. Examples of such applications are fuel, paint, and hazardous material storage areas.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class, and symbol number as required (including title, number, and date of the applicable specification sheet) (see 1.2).

c. When first article inspection is required (see 3.2).

d. Requirement for prohibited materials verification (see 3.4.2. and 4.8.24).

e. Whether 100 percent nonmagnetic material is required (see 3.4.8).

f. Whether watertight fixtures are required (see 3.5.11).

g. Whether the type III ambient temperature range is different than -13 to 140 °F (-25 to 60 °C) (see 3.5.17).

h. If lamps are required for incandescent fixtures (see 3.6.2).

i. Type of starter required if other than specified in the applicable specification sheet (see 3.6.6).

j. Whether switches are required (see 3.6.10).

k. If electromagnetic interference requirements are required (see 3.6.14 and 6.6).

l. Power interface requirements other than specified (see 3.6.18).

m. Whether color filter tubes for fluorescent lamps are required (see 3.7.9.1).

n. Whether increased lifetime is required for type III luminaires (see 3.7.11 and 3.14).

o. Whether lumen projection and measurement data should be submitted to the acquisition activity (see 3.7.11).

p. If high temperature fluorescent lighting fixtures are required (see 3.11).

q. Whether increased energy efficiency is required for type III luminaires (see 3.13).

r. If comparison inspection is required (see 4.3.4).

s. If the source of light should be standard (see 4.8.4).

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t. Test procedure for measuring light output other than IES guides for photometric and colorimetric testing (see 4.8.6).

u. Hazardous location type (see 4.8.15.1).

v. Orientation for type III lumen maintenance test (see 4.8.20.1).

w. Packaging requirements (see 5.1).

x. Acquisition activity and technical authority approval of first article inspection, conformance inspection, projections, and data analysis (see 6.4).

6.3 Provisioning. Provisioning technical documentation, spare parts, and repair parts should be furnished as specified in the ordering requirements.

6.3.1 Ordering of parts. When ordering spare parts or repair parts for the equipment covered by this specification, the ordering requirements should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.4 First article inspection, conformance inspection, projections, and data analysis.

6.4.1 Approval of the first article test procedures and report. First article test procedures and test reports should be submitted for approval by the fixture design activities to the acquisition activity (Naval Shipyards; Private Shipyards/Supervisor of Shipbuilding, Conversion and Repair; and Defense Logistics Agency). These first article test procedures and reports should then be submitted by the acquisition activity to the Technical Authority for approval. The Technical Authority will (by written notice to the acquisition activity) approve, conditionally approve, or disapprove the first article test procedure or report. The acquisition activity should forward the first article inspection report to the Technical Authority for resolution on acceptance or nonacceptance of marginal or nonconforming supplies.

6.4.1.1 Design support information. Any projections and analysis required by this specification should be submitted to the Technical Authority by the acquisition activity; for example, the lumen maintenance projections required by 3.11 and chromaticity required by 3.7.4.

6.4.1.2 Drawings acceptance. A prerequisite to submission of the first article for inspection is evidence satisfactory to the acquisition activity that the Technical Authority has reviewed and accepted the manufacturing drawings for the fixture, light, or part to be furnished under this specification (see appendix A). Drawings should be in AutoCAD, or other approved format, and include physical dimensions, parts lists, and wiring diagrams.

6.4.1.3 Waiving of first article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirements for first article inspection to those bidders offering a product which has previously passed first article inspection. Bidders offering such products, who wish to rely on such test results, must furnish such evidence with the bid.

6.4.2 Previous testing. Unless otherwise directed by the Technical Authority, substitution of comparison inspection in lieu of first article inspection is acceptable and desirable for those companies offering products which have been previously acquired or tested by the Government as long as the requirements of this specification are met.

6.4.3 Testing laboratories. All first article and conformance inspection should be performed at independent laboratories separate from contractor facilities. Testing may be performed at the vendor's testing laboratory provided the laboratory is acceptable to the Technical Authority. Government representatives have the option to witness all testing at independent and vendor laboratories unless otherwise directed by the Technical Authority.

6.5 Comparison inspection. Comparison inspection is not mandatory but is highly desired to assure that production runs conform to the test results of the first article inspection. Comparison inspection should be mandatory for those companies manufacturing the fixtures, lights, or parts under this specification for the first time.

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6.6 Electromagnetic interference. Fluorescent lighting fixtures (type I) manufactured in accordance with previous revisions of this specification met the radiated electromagnetic interference requirements. When more stringent requirements are required (see 6.2), these requirements should be as specified in the purchasing documents and the fixtures modified and tested accordingly. Suggested modification methods may be obtained from the Technical Authority.

6.7 Technical Authority approval and direction. Deviations from specified materials, procedures, and requirements and selection of specific alternative materials and procedures require Technical Authority approval or direction. Requests should include supporting documentation.

6.8 Definitions.

6.8.1 Driver. An LED driver is a power source that uses ship's power as an input and adjusts the output voltage and current to LEDs. A driver can range from something as simple as a resistor to as complex as a constant current power supply.

6.8.2 Excessive corrosion. Excessive corrosion is defined as that which interferes with the electrical or mechanical performance or, in the case of plated metals, corrosion that has passed through the plating and attacked the base metal.

6.8.3 Lamp. As defined in ANSI/IES LS-1-21, Lighting Science: Nomenclature and Definitions for Illuminating Engineering, lamp is a generic term for a manufactured source created to produce optical radiation. Note: Through popular usage, a portable luminaire consisting of a lamp with shade, reflector, enclosing globe, housing, or other accessories is also called a lamp. In such cases, in order to distinguish between the assembled unit and the light source within it, the latter is often called a bulb or tube, if it is electrically powered.

6.8.4 Leakage current. Leakage current is defined as the peak value of current flowing through the connection when any line of the power source is connected to the frame of the fixture for test purposes with the fixture operating.

6.8.5 Luminaire or light fixture. As defined in ANSI/IES LS-1-21, Lighting Science: Nomenclature and Definitions for Illuminating Engineering, a luminaire is a complete lighting unit consisting of a light source(s) and ballast(s) or driver(s) (when applicable), together with the parts designed to distribute the light, to position and protect the light source(s), and to connect the light source(s) to the power supply.

6.8.6 Solid state lighting (SSL). SSL utilizes LEDs, organic LEDs, or polymer LEDs as sources of illumination rather than electrical filaments or gas. The term "solid state" refers to light emitted from a block of solid semiconductor compared to a glass sphere or tube. The physical design of SSL systems provides a greater resistance to shock and vibration. In addition, the use of LEDs significantly increases the luminaires' rated life.

6.8.7 Special tools. Special tools are defined as those tools not listed in the GSA Global Supply Catalog. Copies of this catalog may be obtained from the office of the Defense Contract Management Agency.

6.8.8 Standard stock parts. Standard stock is defined as material listed in the GSA Global Supply Catalog and includes such items as globes, lenses, roundels, lamps, lampholders, cable, cord, wire, bolts, screws, nuts, and washers.

6.8.9 Technical Authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05S may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

6.8.10 TMP_{LED}. A physical temperature measurement, taken at a point designated by the LED chip, package, array, or module manufacturer for their products which act as proxy points for measuring junction temperature.

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6.9 Lumen maintenance predictions for type III luminaires and components. Previous versions of MIL-DTL-16377 (including the unpublished solid state supplement) provided test methods with data extrapolation that resulted in predictions of at least 50,000 hours for 70 percent lumen maintenance. In order to more closely align with the available IES lumen maintenance prediction methodology, the minimum hours to 70 percent lumen maintenance has been changed to 36,000 hours. This is based on the typical 6,000 hours of test data available from the LED chip manufacturers and the TM-21/TM-28 limit of projecting lifetime to no greater than 6 times the available test data. Procuring activities should consider the lumen maintenance of luminaires and components predicted utilizing current IES methodologies to be equivalent to lumen maintenance predictions of 50,000 hours or greater approved under earlier versions of MIL-DTL-16377 and its supplements.

6.10 Subject term (key word) listing.

Explosionproof

Flammability

Fluorescent

IES

Illuminating Engineering Society

Illumination

Incandescent

LED

Light emitting diode

Solid state lighting

SSL

6.11 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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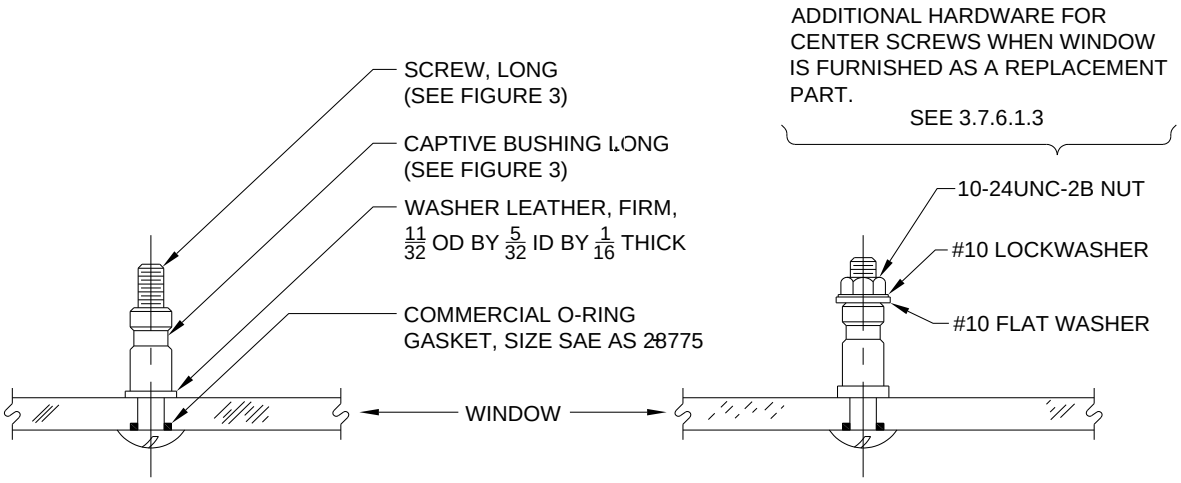


FIGURE 1. Securing screw assembly (long) for watertight window assemblies.

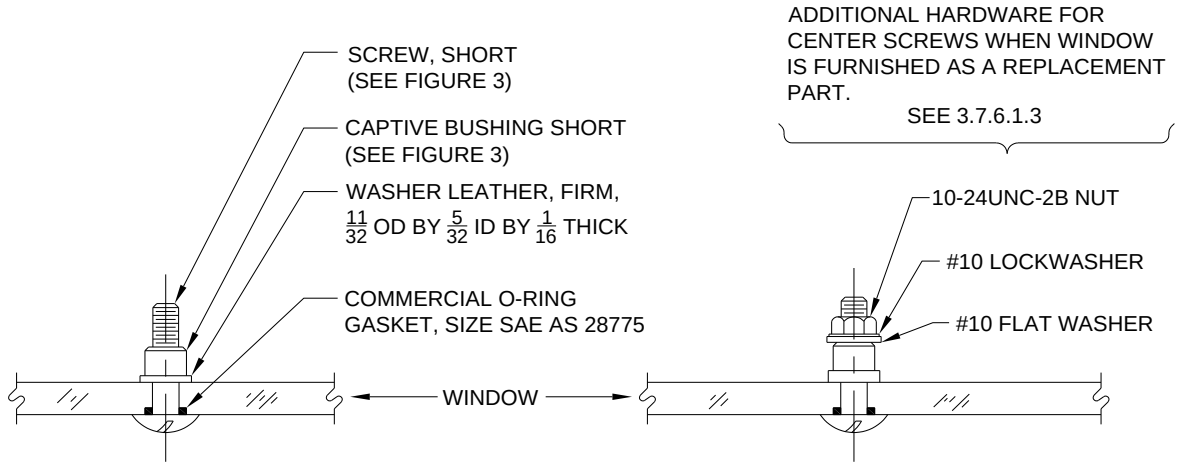
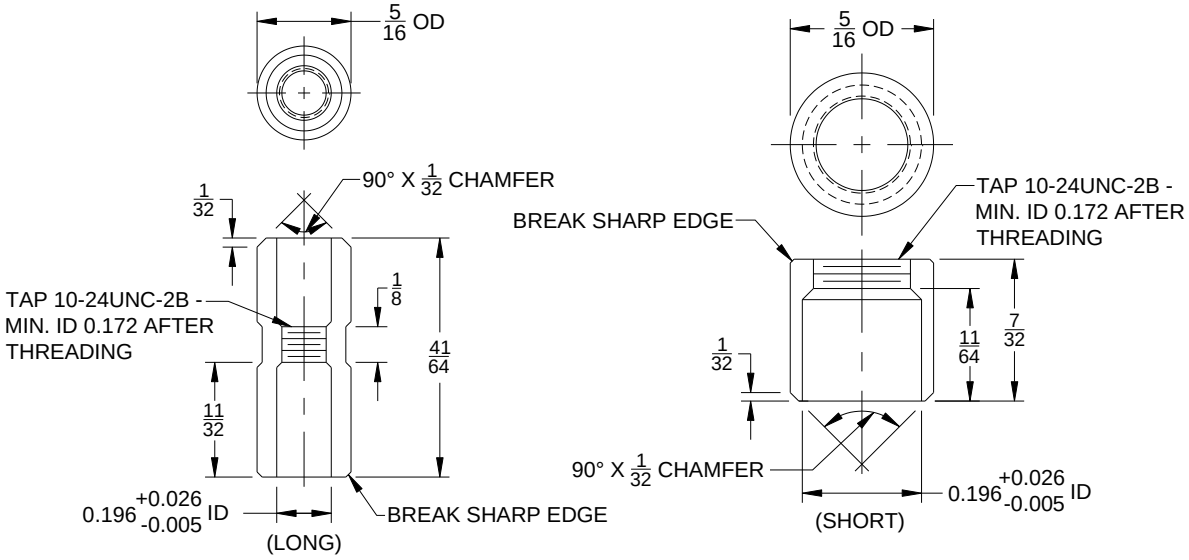
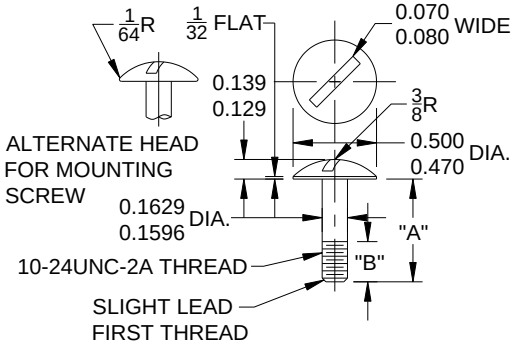


FIGURE 2. Securing screw assembly (short) for watertight window assemblies.

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(A) - CAPTIVE BUSHINGS



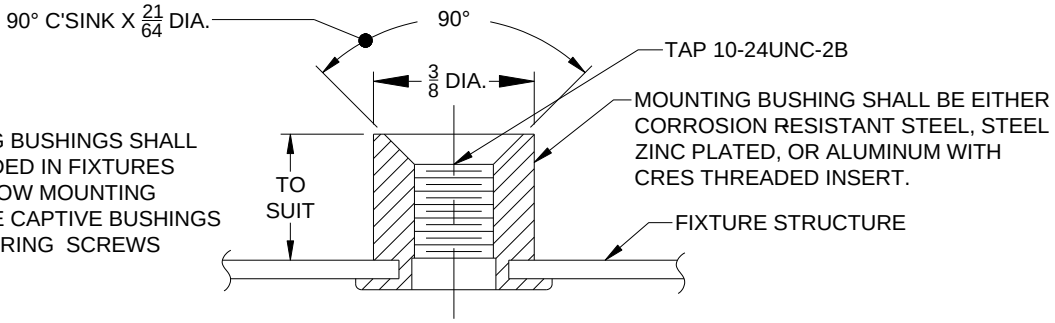
(B) - SECURING SCREWS

MATERIAL:

CAPTIVE BUSHINGS SHALL BE BRASS, NICKEL PLATED, OR ALUMINUM.

SCREWS SHALL BE BRASS, NICKEL PLATED, OR CORROSION RESISTANT STEEL.

USED WITH	"A"	"B"
FIXTURE 1 (LONG)	1 1/4	3/8
FIXTURE 2 (SHORT)	13/16	3/8



NOTE:
MOUNTING BUSHINGS SHALL BE PROVIDED IN FIXTURES FOR WINDOW MOUNTING USING THE CAPTIVE BUSHINGS AND SECURING SCREWS SHOWN.

(C) - MOUNTING BUSHING

FIGURE 3. Captive bushings, securing screws, and mounting bushings.

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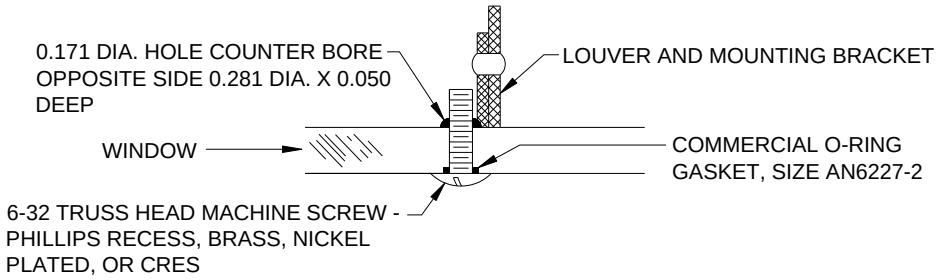


FIGURE 4. Louver attachment for watertight window assemblies.

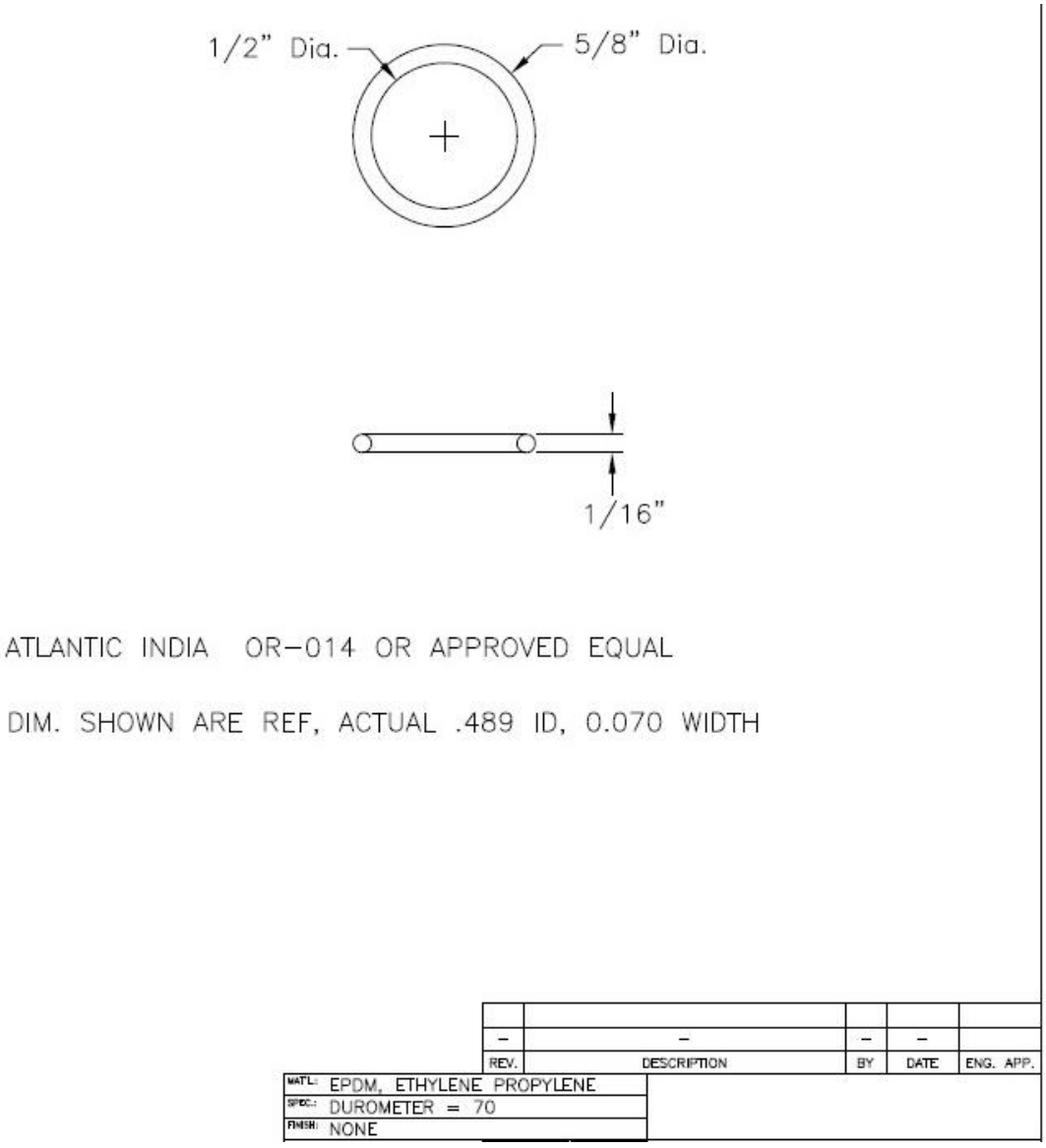


FIGURE 5. Starter washer.

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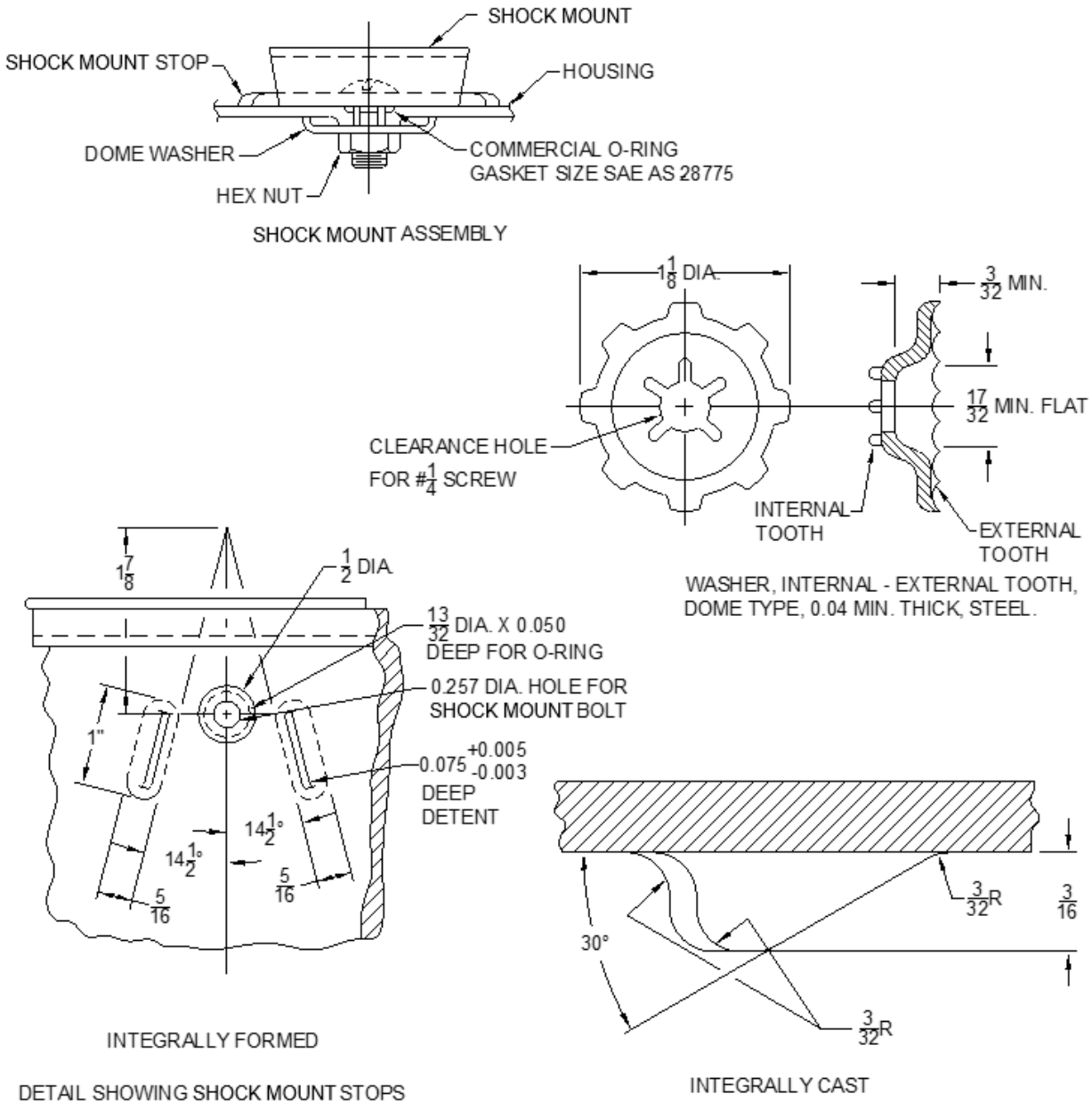
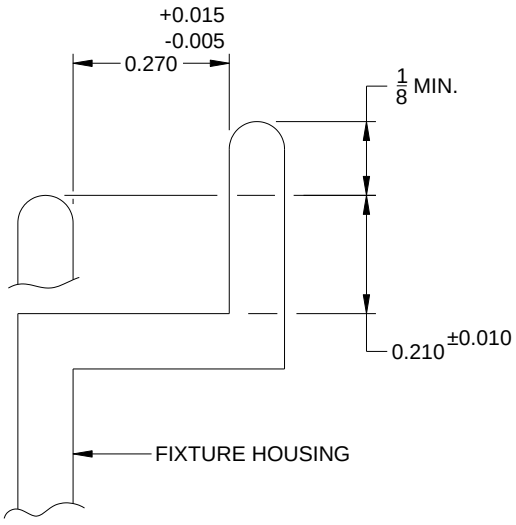
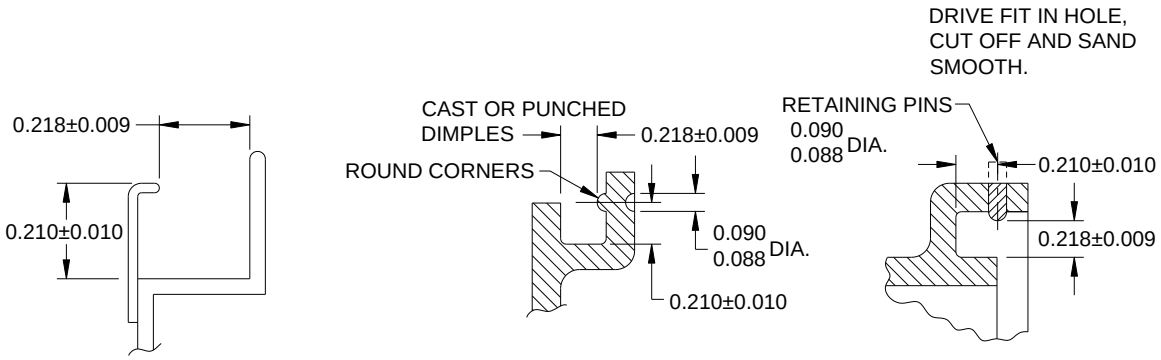


FIGURE 6. Shock mount installation for type I, class 2 fixture.

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DIMENSIONS FOR GASKET GROOVE



(A) CONTINUOUS RETAINER (B) DIMPLES (C) PINS

NOTES:
1. CONTINUOUS GASKET RETAINER SHALL SPAN AT LEAST THE LENGTH SHOWN ON APPLICABLE SPECIFICATION SHEET BY THE TWO (2) EXTREME NOTCHES ON EACH SIDE OF THE WINDOW ASSEMBLIES.
2. LOCATION OF GASKET RETAINING DIMPLES OR PINS ON FIXTURE HOUSINGS SHALL CORRESPOND TO THE LOCATION OF THE NOTCHES SHOWN ON APPLICABLE SPECIFICATION SHEET FOR THE WINDOW ASSEMBLIES.

METHODS OF RETAINING O-RING GASKET

FIGURE 7. Dimensions for gasket groove and methods for retaining O-ring gasket for type I, class 2 fixtures.

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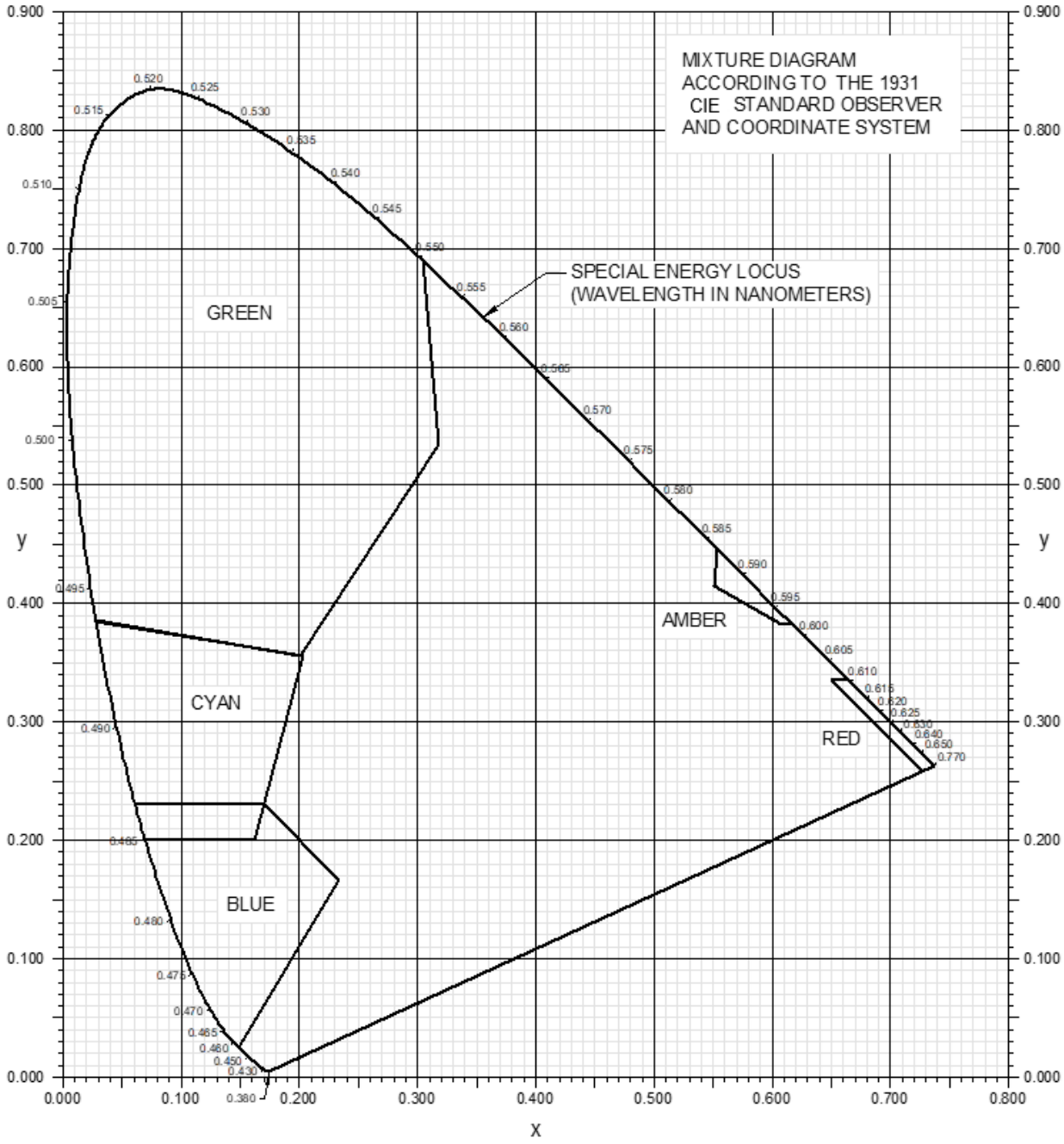


FIGURE 8. Chromaticity diagram illuminant A.

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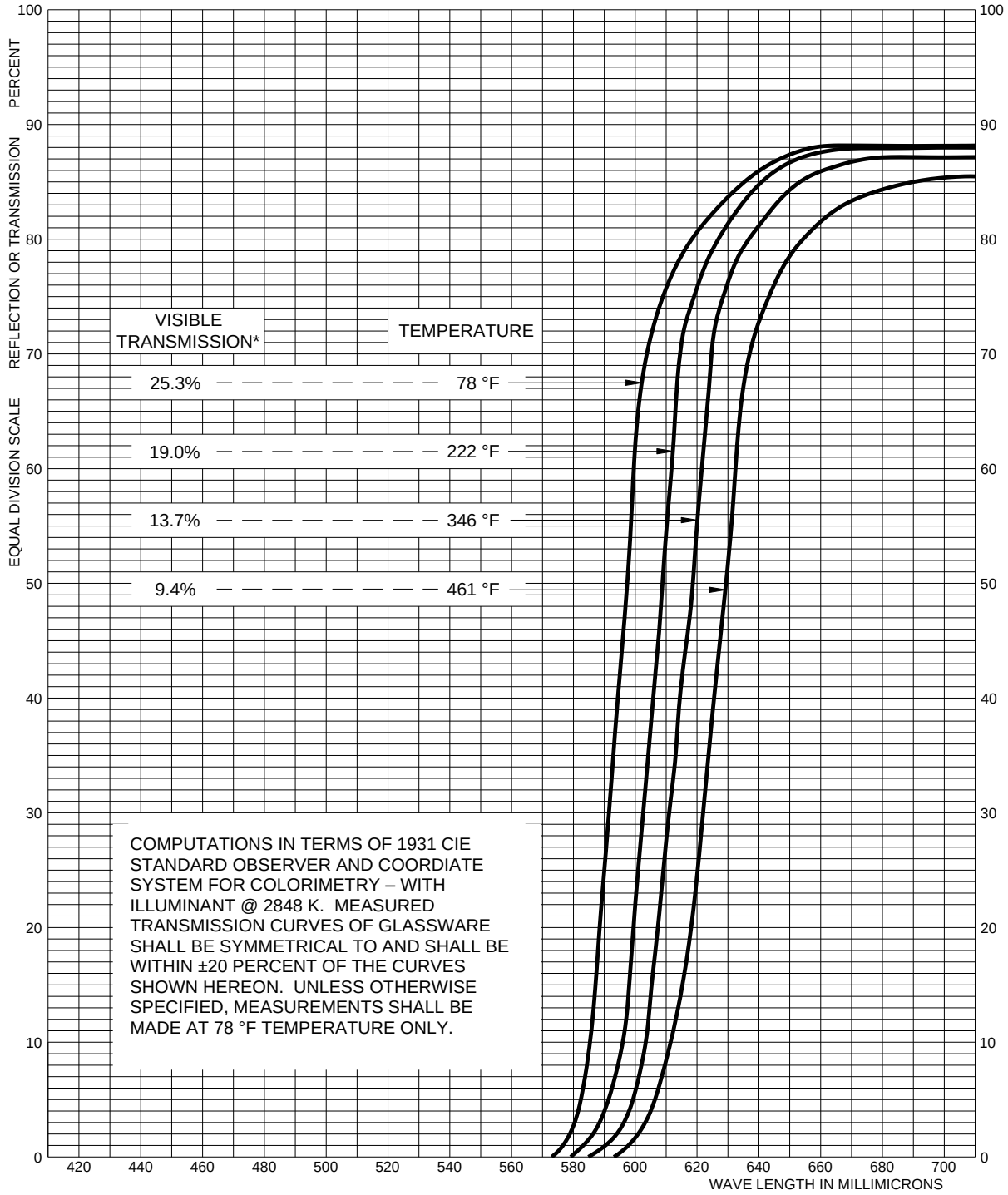


FIGURE 9. Light transmission curves of red glassware.

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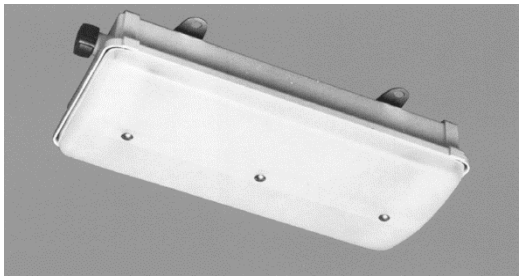


Illustration A: Standard Mount, White Diffusing Window

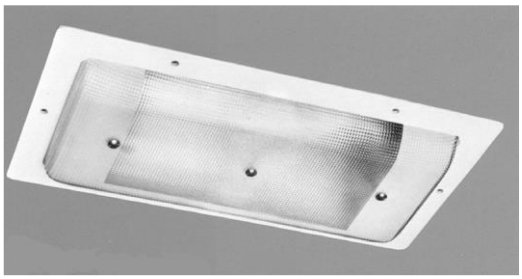


Illustration B: Flush Mount, Clear and White Prismatic Window

FIGURE 10. Typical styles of fluorescent lighting fixtures for general overhead lighting.

MIL-DTL-16377/21A
SYM. NO. 90.2
NSN 6210-00-548-0081
JOHN DOE CO. TROY, NY

EXAMPLE
FOR EQUIPMENT IDENTIFICATION

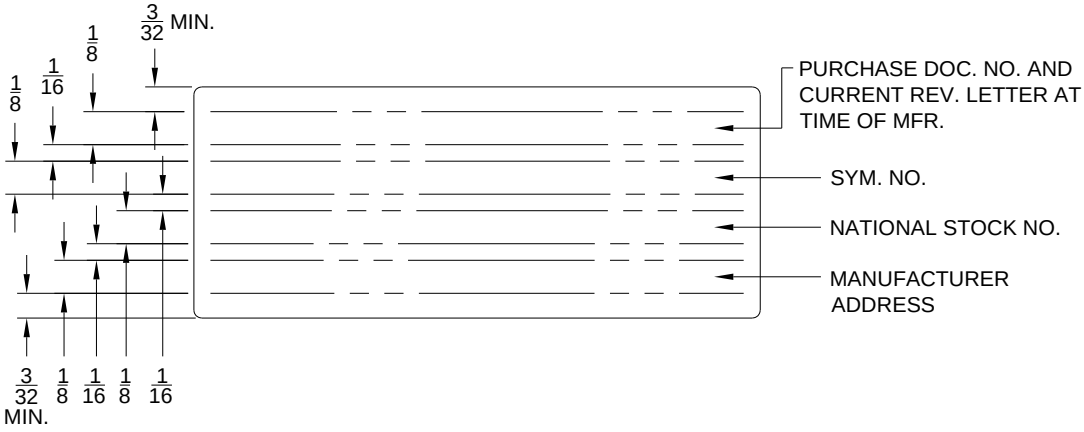
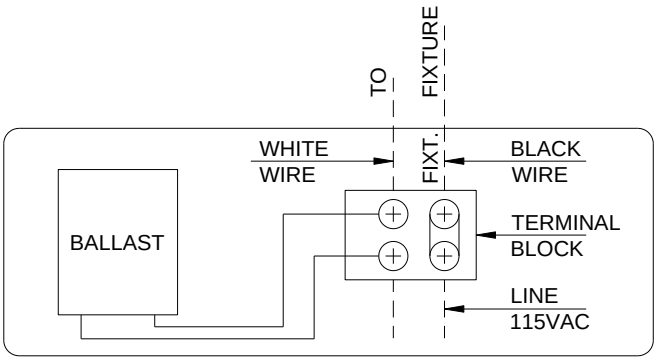


FIGURE 11. Type I labels.

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FOR SINGLE WHITE OR RED CIRCUIT: CONNECT WHITE LEAD TO WHITE LINE TERMINAL. CONNECT RED AND BLACK LEADS TO OTHER LINE TERMINAL.
FOR COMBINATION RED AND WHITE CIRCUITS:
CONNECT WHITE LEAD TO WHITE LINE TERMINAL.
CONNECT BLACK LEAD TO BLACK LINE TERMINAL.
CONNECT RED LEAD TO RED LINE TERMINAL.

FIGURE 12. Type II labels.



TYPICAL WIRING DIAGRAM LABEL

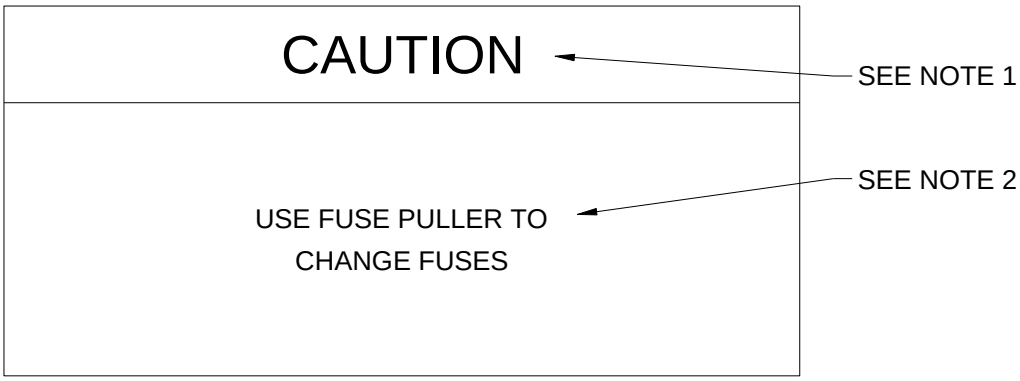
FIGURE 13. Type III labels.

INSTALL THE FOLLOWING LAMP ONLY
NSN 6240-00-246-5054

TYPICAL INFORMATION LABEL

FIGURE 14. Type IV labels.

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NOTES:
1. YELLOW WORDING ON BLACK.
2. BLACK WORDING ON YELLOW.
3. MINIMUM HEIGHT OF LETTERS SHALL BE $\frac{3}{32}$ INCH.

TYPICAL WARNING LABEL

FIGURE 15. Type V labels.

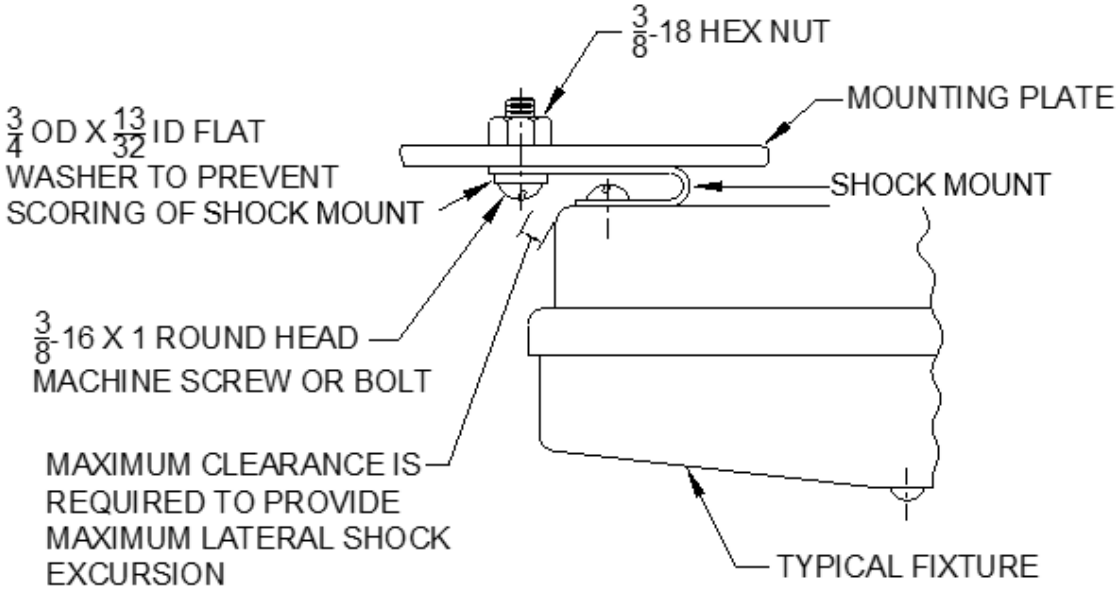
NOTICE TO INSTALLER

BEFORE INSTALLING WINDOW ASSEMBLY TO FIXTURE REMOVE NUT, FLATWASHER, AND LOCKWASHER FROM CENTER SECURING SCREW(S) WHEN FIXTURE PROVIDES FOR CENTER MOUNTING. ASSEMBLE WINDOW TO FIXTURE ONLY WITH SCREWS FURNISHED. NO ADDITIONAL SCREWS ARE NECESSARY. DISCARD LABEL AT ASSEMBLY.

TYPICAL INFORMATION LABEL FOR WINDOWS THAT ARE FURNISHED AS REPAIR PARTS.

FIGURE 16. Type VI labels.

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(A) MOUNTING OF LIGHTS THAT ARE FURNISHED WITH SHOCK MOUNT

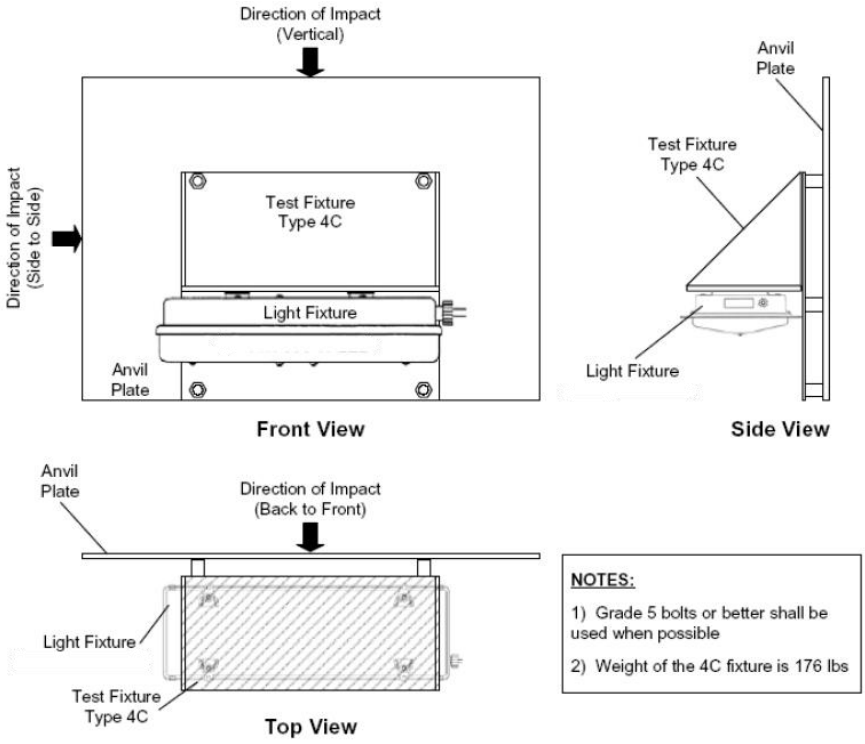


FIGURE 17. Standard methods of mounting lights for shock tests.

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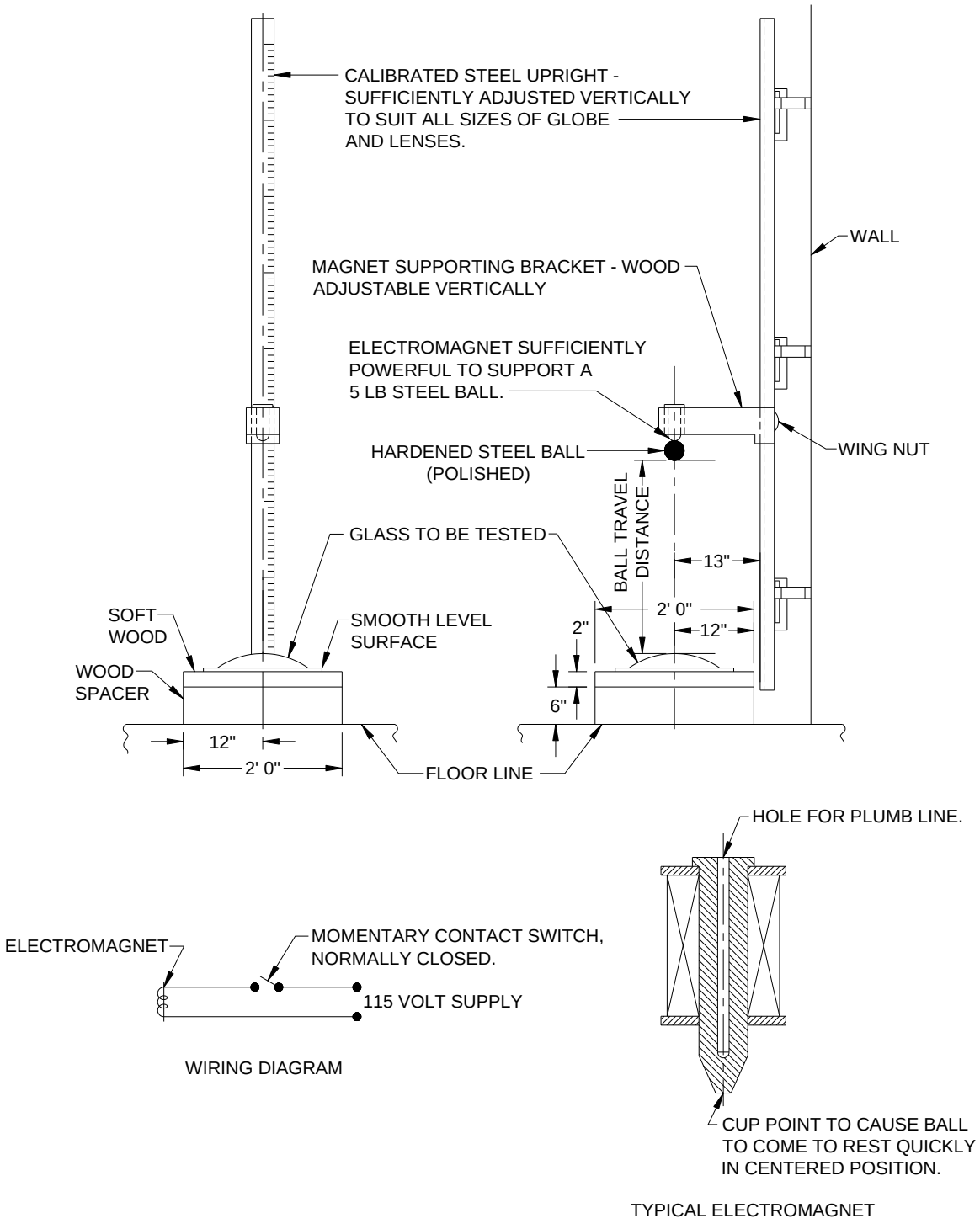
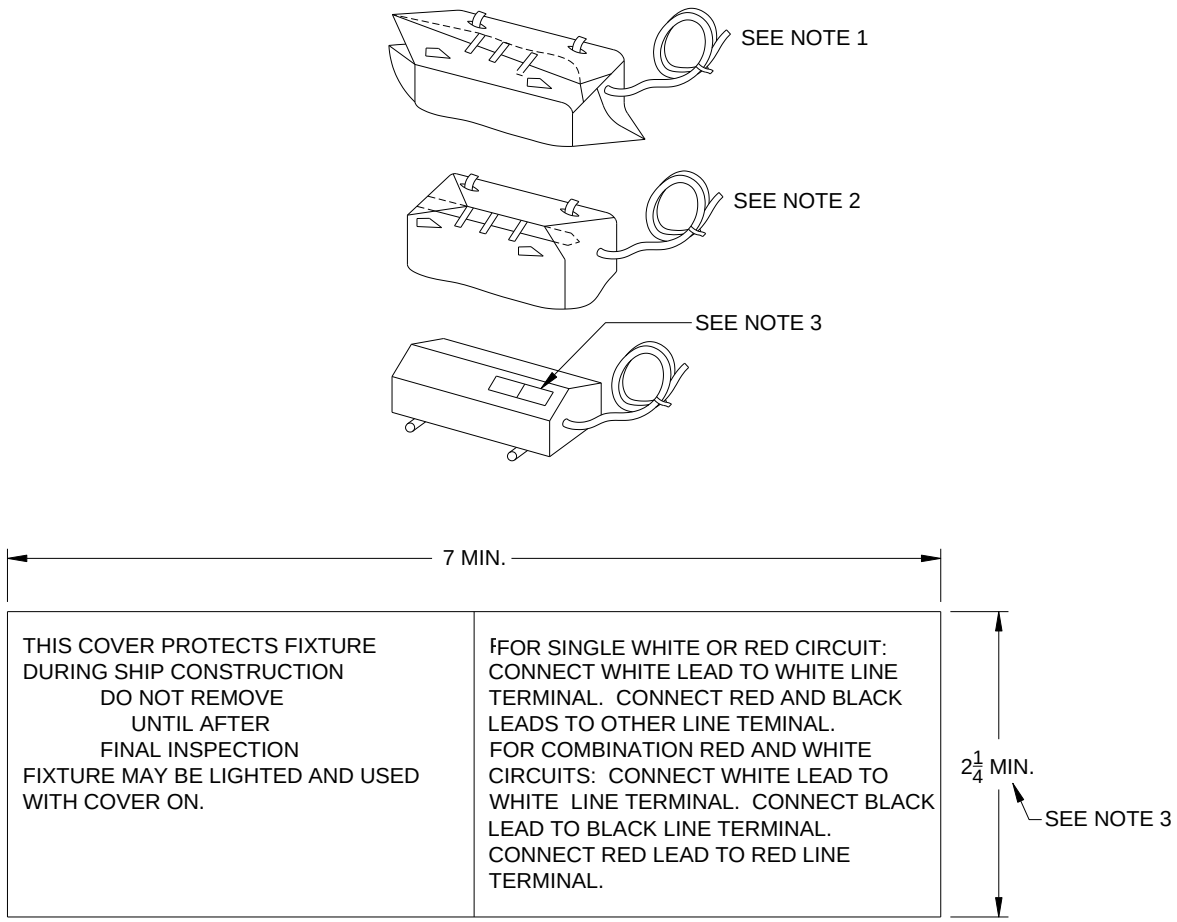


FIGURE 18. Typical ball impact equipment for shock test of glassware.

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NOTES:
1. INSERT FIXTURE IN COVER. SECURE WITH ADHESIVE TAPE OR WELD.
2. FOLD ENDS OVER TOP OF FIXTURE AND SECURE WITH ADHESIVE TAPE OR WELD.
3. THE LEGEND SHOWN ON THE ABOVE DIAGRAM SHALL BE STAMPED ON THE LOWER EDGE OF THE CLOSED SIDE OF THE COVER. SIZE AND STYLE OF TYPE USED SHALL PRODUCE A LEGIBLE COPY WHEN APPLIED TO THE COVER. LEGEND SHALL FILL THE SPACE INDICATED BY THE MINIMUM DIMENSIONS SHOWN ON THE DIAGRAM.

FIGURE 19. Method of installing cover on type I, class 2 fixtures and lights with standard mounting.

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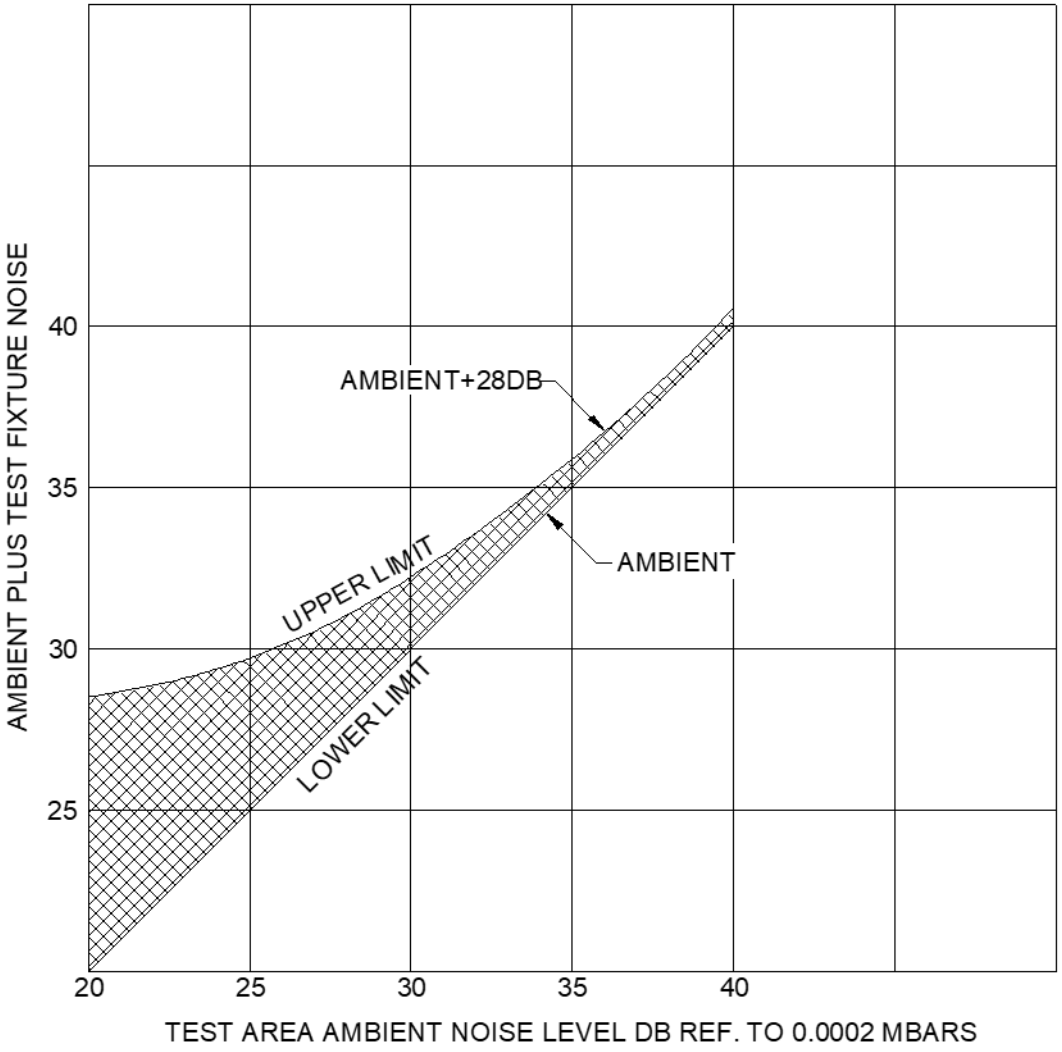
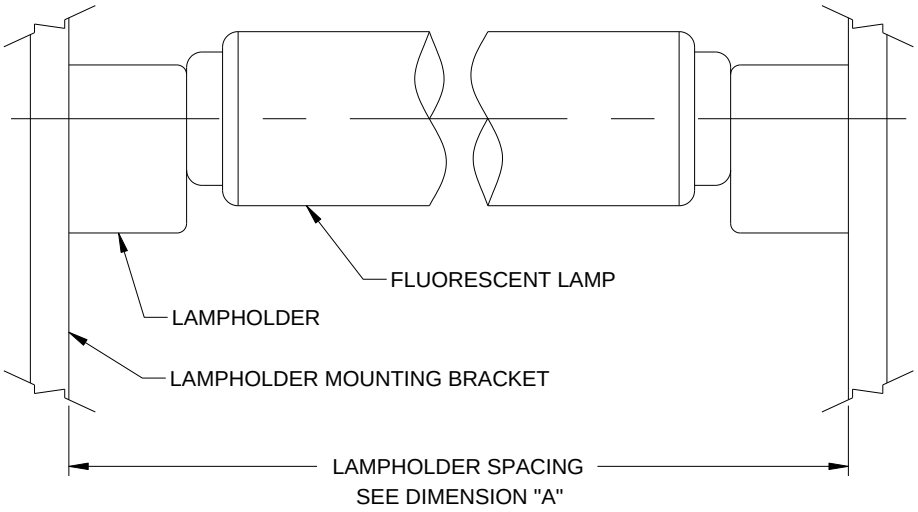


FIGURE 20. Airborne noise acceptable levels.

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Lamp (Watts)	Dimension “A” (Inches) ±0.015
6	9.360
8	12.360
15	18.150
20	24.150

FIGURE 21. Lampholder spacing.

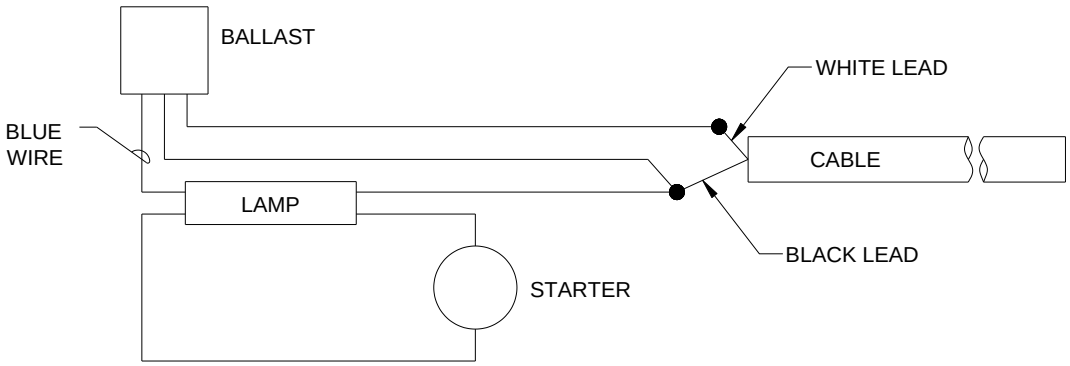


FIGURE 22. Wiring diagram to operate one fluorescent lamp.

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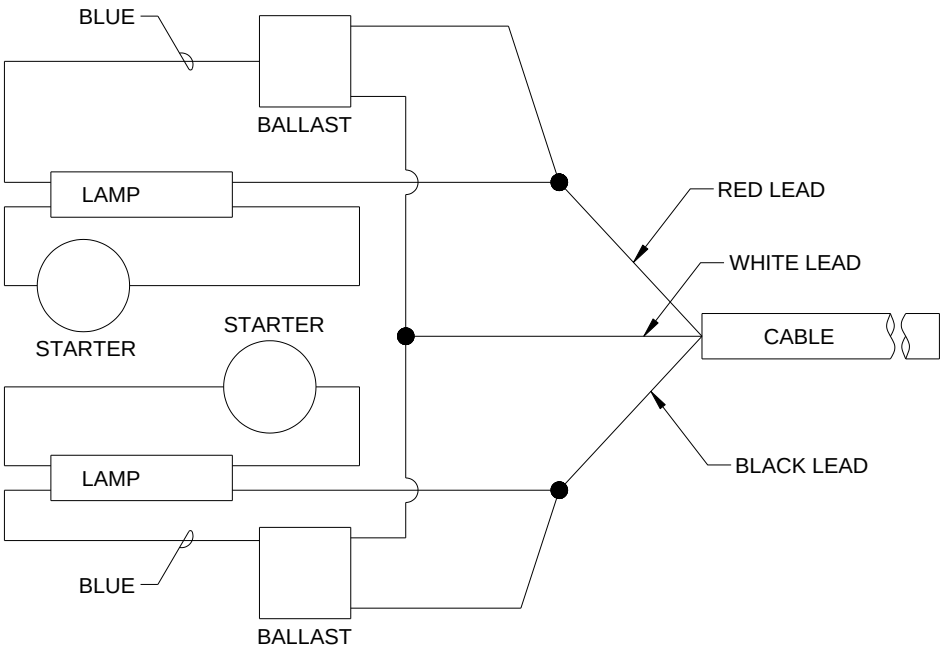


FIGURE 23. Wiring diagram to operate two fluorescent lamps.

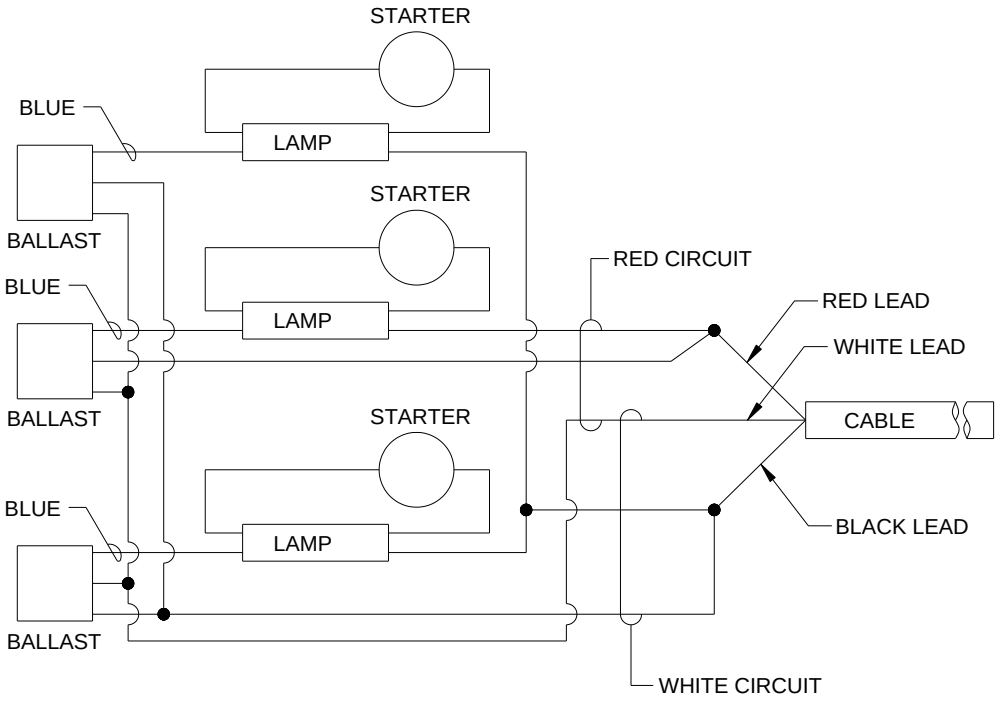
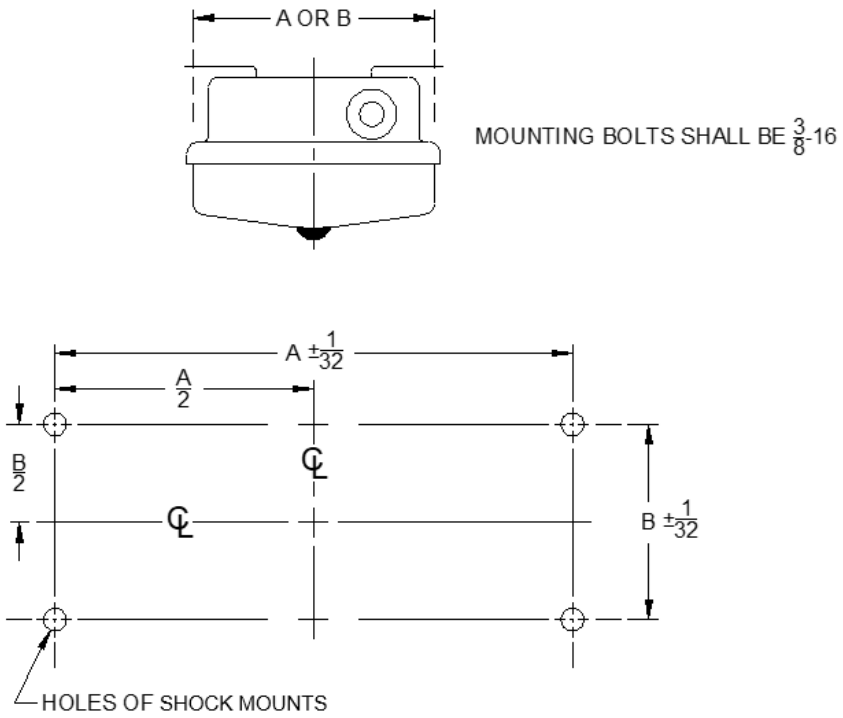


FIGURE 24. Wiring diagram to operate three fluorescent lamps.

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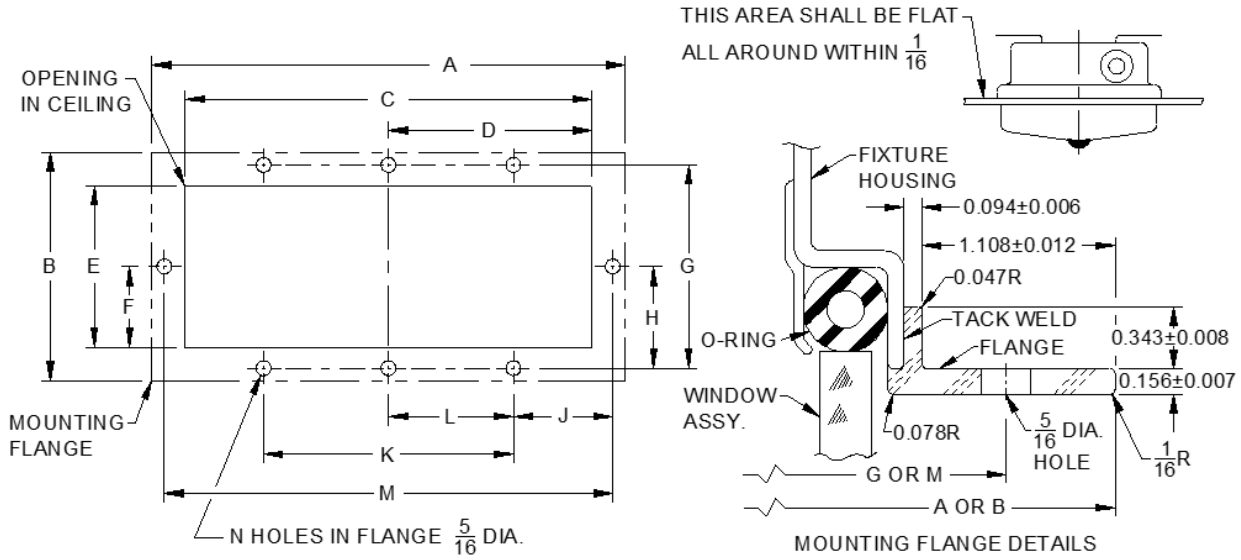
Specification Sheet of MIL-DTL-16377	Symbol Number	A	B
/5	338.1 and 339.1	13⅞	4¾
/6 1/	149.4	9¼	7¼
/7	70.3 and 336	9¼	4¾
/8	331.1 and 347.2	19⅞	4¾
/9	73.3 and 346.2	9¼	7¼
/10	145.4	19⅞	7¼
/11	77.4 and 344.2	19⅞	7¼
/12	333.1, 341.1, and 342.2	19⅞	9¼
/14	80.1 and 79.1	9¼	4¾
/65 1/	81 and 345.2	13⅞	7¼
/66 1/	82 and 343.1	13⅞	9¼
/78	331.1T8 and 347.2T8	19⅞	4¾
/79	77.4T8 and 344.2T8	19⅞	7¼
/80	333.1T8, 341.1T8, and 342.2T8	19⅞	9¼

NOTE:

1/ For replacement purposes only.

FIGURE 25. Mounting dimensions for standard mounting of fluorescent fixtures.

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Specification Sheet of MIL-DTL-16377	Symbol Number	A	B	C	D	E	F	G	H	J	K	L	M	N 1/
/5	338.2 and 339.2	25 9/16	7 1/4	23 9/16	11 25/32	5 1/4	2 5/8	6	3	5 7/32	13 7/9	6 15/16	24 5/16	8
/6	149.5	18 11/16	9 15/16	16 11/16	8 11/32	7 15/16	3 31/32	8 11/16	4 11/32	4 3/32	9 1/4	4 5/8	17 7/16	6
/7	70.4 and 336.1	19 29/32	7 1/4	17 29/32	8 61/64	5 1/4	2 5/8	6	3	4 45/64	9 1/4	4 5/8	18 21/32	6
/8	331.2 and 347.3	31 9/16	7 1/4	29 9/16	14 25/32	5 1/4	2 5/8	6	3	5 7/32	19 7/8	9 15/16	30 5/16	8
/9	73.4 and 346.3	18 11/16	9 15/16	16 11/16	8 11/32	7 15/16	3 31/32	8 11/16	4 11/32	4 3/32	9 1/4	4 5/8	17 7/16	6
/10	145.5	31 5/16	9 15/16	29 5/16	14 21/32	7 15/16	3 31/32	8 11/16	4 11/32	5 3/32	19 7/8	9 15/16	30 1/16	8
/11	77.5 and 344.3	31 5/16	9 15/16	29 5/16	14 21/32	7 15/16	3 31/32	8 11/16	4 11/32	5 3/32	19 7/8	9 15/16	30 1/16	8
/12	333.2, 341.2, and 342.3	31 5/8	11 7/8	29 5/8	14 13/16	9 7/8	4 15/16	10 5/8	5 5/16	5 1/4	19 7/8	9 15/16	30 3/8	8
/14	80.2 and 79.2	17 47/64	5 7/32	16 7/8	8 7/16	3 7/32	1 39/64	3 31/32	1 63/64	---	9 15/16	4 31/32	---	4
/65 2/	81.1 and 345.3	25 5/16	9 15/16	23 5/16	11 21/32	7 15/16	3 31/32	8 11/16	4 11/32	5 3/32	13 7/8	6 15/16	24 1/16	8
/66 2/	82.1 and 343.2	25 5/8	11 7/8	23 5/8	11 13/16	9 7/8	4 15/16	10 5/8	5 5/16	5 1/4	13 7/8	6 15/16	24 3/8	8
/78	331.2T8 and 347.3T8	31 9/16	7 1/4	29 9/16	14 25/32	5 1/4	2 5/8	6	3	5 7/32	19 7/8	9 15/16	30 5/16	8
/79	77.5T8 and 344.3T8	31 5/16	9 15/16	29 5/16	14 21/32	7 15/16	3 31/32	8 11/16	4 11/32	5 3/32	19 7/8	9 15/16	30 1/16	8
/80	333.2T8, 341.2T8, and 342.3T8	31 5/8	11 7/8	29 5/8	14 13/16	9 7/8	4 15/16	10 5/8	5 5/16	5 1/4	19 7/8	9 15/16	30 3/8	8

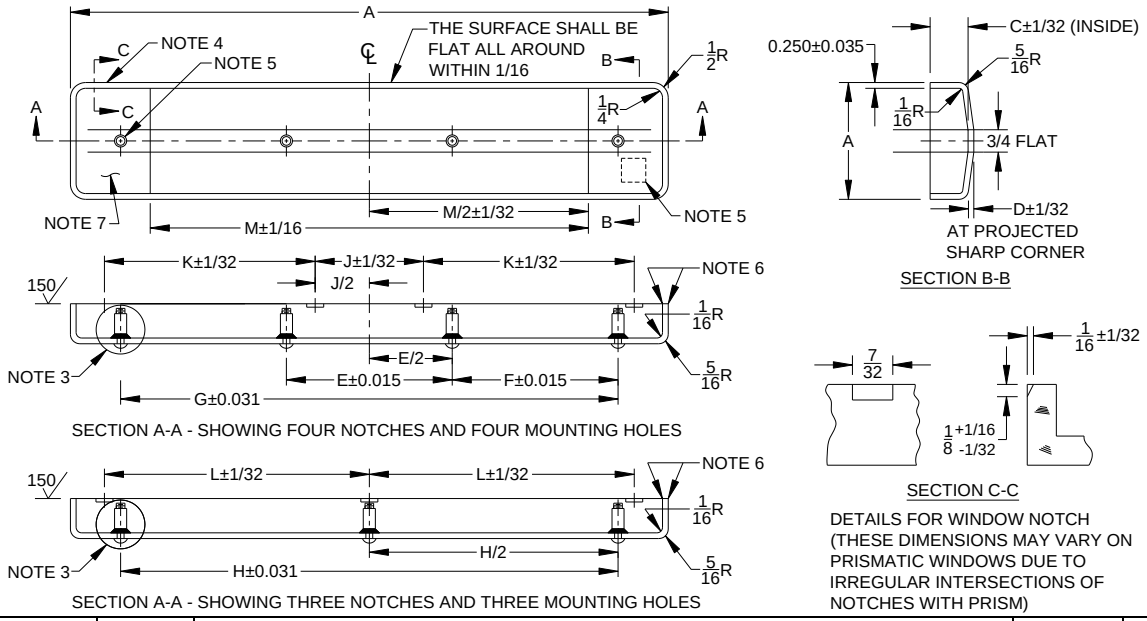
NOTES:

1/ N represents the number of mounting holes in mounting flange.

2/ For replacement purposes only.

FIGURE 26. Mounting dimensions for flush mounting of fluorescent fixtures.

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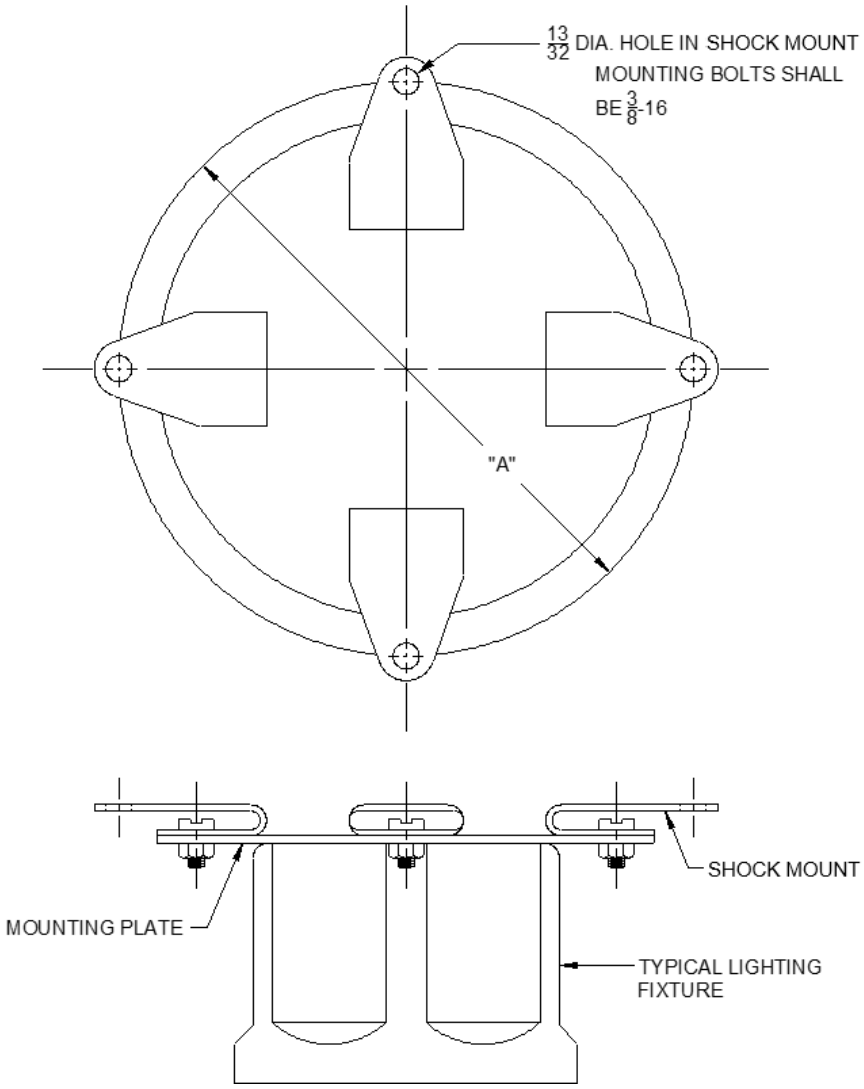
Military Part No.	Material	Dimension												Number of securing screw assemblies	Number of notches each side
		A	B	C	D	E	F	G	H	J	K	L	M		
M16377/5-001	Note 1	22.875	4.593	1 19/64	1/8	---	---	---	19.125	5 41/64	7 21/32	---	17	3	4
M16377/5-002	Note 2	22.797	4.562												
M16377/7-001	Note 1	17.234	4.593	1/2	1/8	---	---	---	13.312	---	---	7 21/32	11	3	3
M16377/7-002	Note 2	17.187	4.562												
M16377/8-001	Note 1	28.875	4.593	1 19/64	1/8	---	---	---	25.125	8 31/32	8 3/4	---	23	3	4
M16377/8-003	Note 2	28.797	4.562												
M16377/9-001	Note 1	16.015	7.265	31/32	7/16	---	---	---	11.782	---	---	7 9/64	11	3	3
M16377/9-003	Note 2	15.968	7.203												
M16377/11-001	Note 1	28.656	7.265	1 1/8	9/16	8.333	8.396	25.125	---	8 3/4	8 3/4	---	23	4	4
M16377/11-003	Note 2	28.563	7.203												
M16377/12-001	Note 1	28.937	9.234	1 31/64	25/32	8.333	8.396	25.125	---	9 1/32	8 3/4	---	23	4	4
M16377/12-004	Note 2	28.844	9.141												
M16377/65-001	Note 1	22.656	7.265	1 1/8	9/16	6.375	6.375	19.125	---	6 3/4	6 3/4	---	17	4	4
M16377/65-002	Note 2	22.563	7.203												
M16377/66-001	Note 1	22.937	9.234	1 31/64	25/32	6.375	6.375	19.125	---	6 3/4	6 3/4	---	17	4	4
M16377/66-002	Note 2	22.844	9.141												

NOTES:

1. Material shall be ¼-inch-thick, white translucent acrylic plastic, type WTLS-4, grade G of MIL-DTL-24191.
2. Material shall be ¼-inch-thick, clear prismatic acrylic plastic, type CLEAR-2 of MIL-DTL-24191. The prisms shall be on the outside surface of the windows.
3. Short window securing screw assemblies in the number indicated on the table shall be installed as specified in 3.7.6.1.3.
4. When required, notches in the number indicated on the table shall be filed as shown on this figure and as specified in 3.7.6.1.4.
5. Drill 0.191-diameter hole and counterbore opposite side 0.323 diameter, 0.050 deep. Break sharp edges on the inside surface on all holes.
6. Break sharp edges 1/64 of an inch around entire window.
7. The inside surface of both ends (bottom, sides, and end) of windows with clear prismatic material shall be finished white as specified in 3.8.4.1.
8. When required, one type VI label shall be applied on the outside surface of the white translucent windows and on the inside surface of the clear and white prismatic windows as specified in 3.7.6.1.3.

FIGURE 27. Window assemblies.

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Specification Sheet of MIL-DTL-16377	Symbol	"A" DIA.
/20	64.1 and 65	7⅝
/21	89 and 90.2	7⅝
/23	57, 57.1, 69, and 69.1	7⅝
/25	48.2 and 68.2	7⅝
/26	112 and 113	7⅝

FIGURE 28. Mounting dimensions for incandescent fixtures.

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CERTIFICATION/DATA REPORT TECHNICAL CONTENT

A.1 SCOPE

A.1.1 Scope. This appendix covers information that should be included in the certification/data report when specified. This appendix is not a mandatory part of the specification. The information contained herein is intended for guidance only.

A.2 CERTIFICATION CONTENT

A.2.1 Certification of drawings acceptance. A prerequisite to submission of the first article for inspection should be provision of satisfactory evidence to the acquisition activity that the Technical Authority (see 6.8.9) has reviewed and accepted the manufacturing drawings for the fixture, light, or part to be furnished under this specification. The supplier should provide drawing acceptance procedures, drawings, and certification data sheets in accordance with the data ordering documents included in the contract or order, and as specified in A.2.1.1 through A.2.1.1.2.

A.2.1.1 Drawing acceptance procedures. Two sets of prints of the manufacturing drawings for the fixture, light, or part, followed by two sets of prints of the certification data sheet should be provided for review. Acceptance should be based upon the completeness of the drawing and other data submitted conclusively providing proof of proper performance of the fixture, light, or part in accordance with this specification and the applicable specification sheet. Drawing acceptance should be obtained in writing.

A.2.1.1.1 Submission of revised drawings. When a drawing is not accepted, submission of a revised drawing should be accomplished. In addition, the Technical Authority may require other drawings and documents for reference.

A.2.1.1.2 Changes. There should be no changes made in any accepted drawing, as they will require re-approval by the Technical Authority. If changes are required, revised drawings should be submitted as specified in A.2.1.1.

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CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 6210-2019-021)

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